CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2018 Examination | **Paper Code:** 18024 | **Complete 100% Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION – A (VERY SHORT ANSWER QUESTIONS • 3 MARKS EACH • ALL 5 COMPULSORY)

Q1. Obtain a real root of the equation $f(x) = x^3 + x^2 - 1 = 0$ using Bisection method correct to 3 decimal places. [3 Marks]

Solution:

Let $f(x) = x^3 + x^2 - 1$.

- For $x = 0$: $f(0) = 0 + 0 - 1 = -1 < 0$
- For $x = 1$: $f(1) = 1 + 1 - 1 = +1 > 0$

Since $f(0) \cdot f(1) < 0$, a real root lies in the open interval **(0, 1)**.

n	a	b	c = (a+b)/2	f(c)	Sign	Updated Interval
1	0.0000	1.0000	0.5000	-0.6250	Negative (< 0)	[0.5000, 1.0000]
2	0.5000	1.0000	0.7500	-0.0156	Negative (< 0)	[0.7500, 1.0000]
3	0.7500	1.0000	0.8750	+0.4355	Positive (> 0)	[0.7500, 0.8750]
4	0.7500	0.8750	0.8125	+0.1965	Positive (> 0)	[0.7500, 0.8125]
5	0.7500	0.8125	0.7812	+0.0872	Positive (> 0)	[0.7500, 0.7812]
6	0.7500	0.7812	0.7656	+0.0350	Positive (> 0)	[0.7500, 0.7656]
7	0.7500	0.7656	0.7578	+0.0095	Positive (> 0)	[0.7500, 0.7578]
8	0.7500	0.7578	0.7539	-0.0031	Negative (< 0)	[0.7539, 0.7578]
9	0.7539	0.7578	0.7559	+0.0032	Positive (> 0)	[0.7539, 0.7559]
10	0.7539	0.7559	0.7549	+0.00002	Positive (≥ 0)	[0.7539, 0.7549]
11	0.7539	0.7549	0.7544	-0.0015	Negative (< 0)	[0.7544, 0.7549]

At iteration 11, interval width is $|0.7549 - 0.7544| = 0.0005 < 0.001$.

Final Converged Root: x ≈ 0.755 (correct to 3 decimal places)

Q2. Define backward differences and prove the operator identity $E\nabla = \nabla E = \Delta$. [3 Marks]

Solution:

Definition: For equispaced points $x_i = x_0 + i \cdot h$, the **Backward Difference Operator (∇)** is defined as:

$$\nabla y_i = y_i - y_{i-1} \quad \text{and} \quad \nabla^k y_i = \nabla^{k-1} y_i - \nabla^{k-1} y_{i-1} \ (k \geq 2)$$

Backward Difference Table for ordered pairs (x_0, y_0) to (x_4, y_4):

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0	y_0	–	–	–	–
x_1	y_1	$\nabla y_1 = y_1 - y_0$	–	–	–
x_2	y_2	$\nabla y_2 = y_2 - y_1$	$\nabla^2 y_2$	–	–
x_3	y_3	$\nabla y_3 = y_3 - y_2$	$\nabla^2 y_3$	$\nabla^3 y_3$	–
x_4	y_4	$\nabla y_4 = y_4 - y_3$	$\nabla^2 y_4$	$\nabla^3 y_4$	$\nabla^4 y_4$

Operator Proofs:

1. $E \bullet \nabla f(x) = E\,[f(x) - f(x-h)] = f(x+h) - f(x) = \Delta f(x) \Rightarrow \boldsymbol{E\nabla = \Delta}$
2. $\nabla \bullet E\, f(x) = \nabla\,[f(x+h)] = f(x+h) - f(x) = \Delta f(x) \Rightarrow \boldsymbol{\nabla E = \Delta}$

Hence Proved: $E\nabla = \nabla E = \Delta$

Q3. Find the value of $\frac{dy}{dx}$ at $x = 1.2$ for points (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552)... [3 Marks]

Solution:

Given step size $h = 1.2 - 1.0 = 0.2$. Let $x_0 = 1.2$, $y_0 = 3.3201$.

Neighbors: $x_{-1} = 1.0$, $y_{-1} = 2.7183$ and $x_1 = 1.4$, $y_1 = 4.0552$.

Central Difference Formula (First Derivative):

$$f'(x_0) \approx \frac{y_1 - y_{-1}}{2h}$$

$$f'(1.2) = \frac{4.0552 - 2.7183}{2(0.2)} = \frac{1.3369}{0.40} = 3.34225$$

(Newton's Forward Difference check: $f'(1.2) \approx (1/0.2)[0.7351 - 0.5(0.1627) + (1/3)(0.0361)] = 3.3289$. True derivative $e^{1.2} \approx 3.3201$.)

dy/dx at x = 1.2 is 3.34225

Q4. Solve the following linear system using Gauss Elimination method: [3 Marks]

$$10x + y + z = 12$$
$$2x + 10y + z = 13$$
$$x + y + 3z = 5$$

Solution:

Step 1: Form the Augmented Matrix [A | B]:

$$\left(\begin{array}{ccc|c} 10 & 1 & 1 & 12 \\ 2 & 10 & 1 & 13 \\ 1 & 1 & 3 & 5 \end{array}\right)$$

Step 2: Forward Elimination to Upper Triangular Form:

- $R_2 \to R_2 - (2/10)R_1 = R_2 - 0.2R_1 \Rightarrow$ [0, 9.8, 0.8 | 10.6]
- $R_3 \to R_3 - (1/10)R_1 = R_3 - 0.1R_1 \Rightarrow$ [0, 0.9, 2.9 | 3.8]
- $R_3 \to R_3 - (0.9 / 9.8)R_2 \Rightarrow$ [0, 0, 2.82653 | 2.82653]

$$\left(\begin{array}{ccc|c} 10 & 1 & 1 & 12 \\ 0 & 9.8 & 0.8 & 10.6 \\ 0 & 0 & 2.82653 & 2.82653 \end{array}\right)$$

Step 3: Back Substitution:

- From Row 3: 2.82653 • z = 2.82653 ⇒ **z = 1**
- From Row 2: 9.8 • y + 0.8(1) = 10.6 ⇒ 9.8 • y = 9.8 ⇒ **y = 1**
- From Row 1: 10 • x + 1(1) + 1(1) = 12 ⇒ 10 • x = 10 ⇒ **x = 1**

Unique Solution: x = 1, y = 1, z = 1

Q5. Solve $\frac{dy}{dx} = x + y^2$, with y(0) = 1 using Picard's method up to second approximation. [3 Marks]

Solution:

Picard's Successive Approximation Formula:

$$y^{(n)}(x) = y_0 + \int_{x_0}^{x} f(t, y^{(n-1)}(t))\, dt$$

Given initial conditions: $x_0 = 0$, $y_0 = 1$, and $f(t, y) = t + y^2$.

First Approximation ($y^{(1)}$):

$$y^{(1)}(x) = 1 + \int_0^x (t + 1^2)\, dt = 1 + \left[\frac{t^2}{2} + t\right]_0^x = 1 + x + \frac{x^2}{2}$$

Second Approximation ($y^{(2)}$):

$$y^{(2)}(x) = 1 + \int_0^x \left[t + (1 + t + \frac{t^2}{2})^2\right] dt$$

Expand: $(1 + t + \frac{t^2}{2})^2 = 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4}$

$$\text{Integrand} = t + 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4} = 1 + 3t + 2t^2 + t^3 + \frac{t^4}{4}$$

$$y^{(2)}(x) = 1 + \left[t + \frac{3t^2}{2} + \frac{2t^3}{3} + \frac{t^4}{4} + \frac{t^5}{20}\right]_0^x$$

$$y^{(2)}(x) = 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{20}x^5$$

$y^{(1)} = 1 + x + x^2/2$ | $y^{(2)} = 1 + x + (3/2)x^2 + (2/3)x^3 + (1/4)x^4 + (1/20)x^5$

SECTION – B (SHORT ANSWER QUESTIONS • 7.5 MARKS EACH • ATTEMPT ANY TWO)

Q6. Find a real root of the equation $2x = \log_{10}(x) + 7$ between 3 and 4 using Method of False Position (Regula Falsi). [7.5 Marks]

Solution:

Let $f(x) = 2x - \log_{10}(x) - 7 = 0$.

- For $x_0 = 3$: $f(3) = 2(3) - \log_{10}(3) - 7 = 6 - 0.47712 - 7 = -1.47712 < 0$
- For $x_1 = 4$: $f(4) = 2(4) - \log_{10}(4) - 7 = 8 - 0.60206 - 7 = +0.39794 > 0$

Since $f(3)$ and $f(4)$ have opposite signs, the root lies in the interval **(3, 4)**.

Regula Falsi Iteration Formula:

$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$

Iteration 1:

$$x_2 = \frac{3(0.39794) - 4(-1.47712)}{0.39794 - (-1.47712)} = \frac{1.19382 + 5.90848}{1.87506} = \frac{7.10230}{1.87506} \approx 3.78777$$

Evaluate $f(x_2)$:

$f(3.78777) = 2(3.78777) - \log_{10}(3.78777) - 7 = 7.57554 - 0.57838 - 7 = -0.00284 < 0$.

Root lies in **[3.78777, 4.00000]**.

Iteration 2:

$$x_3 = \frac{3.78777(0.39794) - 4(-0.00284)}{0.39794 - (-0.00284)} = \frac{1.50731 + 0.01136}{0.40078} = \frac{1.51867}{0.40078} \approx 3.7893$$

Evaluate $f(x_3)$: $f(3.7893) = 2(3.7893) - \log_{10}(3.7893) - 7 = 7.57860 - 0.57856 - 7 = +0.00004 \approx 0$.

Final Converged Root: x ≈ 3.7893

Q7. Find the missing terms in the following table using finite differences: [7.5 Marks]

x	0	5	10	15	20	25	30
y	1	3	? (y_2)	73	225	? (y_5)	1153

Solution:

Let the missing values be y_2 (at $x = 10$) and y_5 (at $x = 25$).

Since 5 values are given (y_0, y_1, y_3, y_4, y_6), the data can be represented by a polynomial of degree 4.

Therefore, the 5th finite difference is identically zero: $\Delta^5 y_0 = 0$ and $\Delta^5 y_1 = 0$.

Using shift operator $\Delta = E - 1$:

$$(E-1)^5 y = (E^5 - 5E^4 + 10E^3 - 10E^2 + 5E - 1)\, y = 0$$

Equation 1 (applying on y_0):

$$y_5 - 5y_4 + 10y_3 - 10y_2 + 5y_1 - y_0 = 0$$
$$y_5 - 5(225) + 10(73) - 10y_2 + 5(3) - 1 = 0$$
$$y_5 - 1125 + 730 - 10y_2 + 15 - 1 = 0$$
$$\mathbf{-10y_2 + y_5 = 381} \qquad \text{— (Equation 1)}$$

Equation 2 (applying on y_1):

$$y_6 - 5y_5 + 10y_4 - 10y_3 + 5y_2 - y_1 = 0$$
$$1153 - 5y_5 + 10(225) - 10(73) + 5y_2 - 3 = 0$$
$$1153 - 5y_5 + 2250 - 730 + 5y_2 - 3 = 0$$
$$5y_2 - 5y_5 + 2670 = 0 \Rightarrow y_2 - y_5 = -534 \Rightarrow \mathbf{y_5 - y_2 = 534} \qquad \text{— (Equation 2)}$$

Solving the Linear System:

From Equation 2: $y_5 = y_2 + 534$.

Substitute into Equation 1: $-10y_2 + (y_2 + 534) = 381 \Rightarrow -9y_2 = 381 - 534 = -153 \Rightarrow \mathbf{y_2 = 17}$.

Then $y_5 = 17 + 534 = \mathbf{551}$.

Missing Values: $y_2 = 17$ (at x = 10) | $y_5 = 551$ (at x = 25)

Q8. Evaluate $I = \int_3^7 x^2 \log_{10}(x)\, dx$ using Simpson's 1/3 rule with $n = 4$. [7.5 Marks]

Solution:

Given limits $a = 3$, $b = 7$, and number of sub-intervals $n = 4$.

Step size: $h = (b - a) / n = (7 - 3) / 4 = \mathbf{1}$.

i	x_i	Function $f(x) = x^2 \log_{10}(x)$	Ordinate y_i
0	3	$3^2 \cdot \log_{10}(3) = 9 \times 0.477121$	4.2941
1	4	$4^2 \cdot \log_{10}(4) = 16 \times 0.602060$	9.6330
2	5	$5^2 \cdot \log_{10}(5) = 25 \times 0.698970$	17.4743
3	6	$6^2 \cdot \log_{10}(6) = 36 \times 0.778151$	28.0134
4	7	$7^2 \cdot \log_{10}(7) = 49 \times 0.845098$	41.4099

Simpson's 1/3 Rule Formula:

$$I = \frac{h}{3}\left[(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2)\right]$$

Sum of Extremes: $y_0 + y_4 = 4.2941 + 41.4099 = \mathbf{45.7040}$

Sum of Odd Ordinates: $4(y_1 + y_3) = 4(9.6330 + 28.0134) = 4(37.6464) = \mathbf{150.5856}$

Sum of Even Ordinates: $2(y_2) = 2(17.4743) = \mathbf{34.9486}$

Total Bracket Sum $= 45.7040 + 150.5856 + 34.9486 = \mathbf{231.2382}$

$$I = \frac{1}{3} \times 231.2382 = \mathbf{77.0794}$$

Evaluated Integral: $I \approx 77.0794$

SECTION – C (LONG ANSWER QUESTIONS • 15 MARKS EACH • ATTEMPT ANY THREE)

Q9. Use Newton-Raphson method to find a real root of $\sin(x) = x / 2$ correct to 3 decimal places between π/2 and π. [15 Marks]

Solution:

Let $f(x) = \sin(x) - \frac{x}{2} = 0$.

Differentiating with respect to x: $f'(x) = \cos(x) - 0.5$.

Given interval: [π/2, π] ≈ [1.5708, 3.1416] radians.

Let initial approximation be $\mathbf{x_0 = 2.0}$ **radians**.

Newton-Raphson Iteration Formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Iteration 1 (at $x_0 = 2.0$):

$$f(2.0) = \sin(2.0) - 1.0 = 0.90930 - 1.0 = \mathbf{-0.09070}$$
$$f'(2.0) = \cos(2.0) - 0.5 = -0.41615 - 0.5 = \mathbf{-0.91615}$$
$$x_1 = 2.0 - \frac{-0.09070}{-0.91615} = 2.0 - 0.09900 = \mathbf{1.90100}$$

Iteration 2 (at $x_1 = 1.90100$):

$$f(1.90100) = \sin(1.90100) - \frac{1.90100}{2} = 0.94598 - 0.95050 = \mathbf{-0.00452}$$
$$f'(1.90100) = \cos(1.90100) - 0.5 = -0.32422 - 0.5 = \mathbf{-0.82422}$$
$$x_2 = 1.90100 - \frac{-0.00452}{-0.82422} = 1.90100 - 0.00548 = \mathbf{1.89552}$$

Iteration 3 (at $x_2 = 1.89552$):

$$f(1.89552) = \sin(1.89552) - \frac{1.89552}{2} = 0.94775 - 0.94776 = \mathbf{-0.00001}$$
$$f'(1.89552) = \cos(1.89552) - 0.5 = -0.31899 - 0.5 = \mathbf{-0.81899}$$
$$x_3 = 1.89552 - \frac{-0.00001}{-0.81899} = 1.89552 - 0.00001 = \mathbf{1.89549}$$

Root: x ≈ 1.895 radians (or ≈ 108.57°)

Q10. Find the Lagrange interpolating polynomial of degree 2 for *y = $\log_e(x)$* from the given table. Hence determine *$\log_e(2.7)$*. [15 Marks]

x	2.0	2.5	3.0
y = $\log_e$(x)	0.69315	0.91629	1.09861

Solution:

Given $x_0 = 2.0, y_0 = 0.69315$; $x_1 = 2.5, y_1 = 0.91629$; $x_2 = 3.0, y_2 = 1.09861$.

Target value: $x = 2.7$.

Lagrange Interpolation Formula for n = 2:

$$P_2(x) = L_0(x)\, y_0 + L_1(x)\, y_1 + L_2(x)\, y_2$$

Calculating Lagrange Cardinal Polynomials at x = 2.7:

$$L_0(2.7) = \frac{(2.7-2.5)(2.7-3.0)}{(2.0-2.5)(2.0-3.0)} = \frac{(0.2)(-0.3)}{(-0.5)(-1.0)} = \frac{-0.06}{0.50} = \mathbf{-0.12}$$
$$L_1(2.7) = \frac{(2.7-2.0)(2.7-3.0)}{(2.5-2.0)(2.5-3.0)} = \frac{(0.7)(-0.3)}{(0.5)(-0.5)} = \frac{-0.21}{-0.25} = \mathbf{+0.84}$$
$$L_2(2.7) = \frac{(2.7-2.0)(2.7-2.5)}{(3.0-2.0)(3.0-2.5)} = \frac{(0.7)(0.2)}{(1.0)(0.5)} = \frac{0.14}{0.50} = \mathbf{+0.28}$$

Verification: $L_0 + L_1 + L_2 = -0.12 + 0.84 + 0.28 = 1.00$ ✓

Evaluating $P_2(2.7)$:

$$P_2(2.7) = (-0.12)(0.69315) + (0.84)(0.91629) + (0.28)(1.09861)$$
$$= -0.083178 + 0.769684 + 0.307611$$
$$= \mathbf{0.994117 \approx 0.99412}$$

(Exact analytical value: ln(2.7) ≈ 0.99325; error ≈ 0.00087.)

Interpolated Value: $\log_e(2.7) \approx 0.99412$

Q11. Use Simpson's 3/8 rule to evaluate $\int_0^1 \frac{dx}{1+x}$ with $h = 1/6$. [15 Marks]

Solution:

Given $a = 0$, $b = 1$, step size $h = 1/6 \Rightarrow n = (1 - 0)/(1/6) = 6$ subintervals.

Integrand: $f(x) = \frac{1}{1+x}$.

i	x_i	Exact $y_i = 1 / (1 + x_i)$	Decimal Value
0	0	1 / 1	1.000000
1	1/6	1 / (7/6) = 6/7	0.857143
2	2/6	1 / (8/6) = 6/8 = 3/4	0.750000
3	3/6	1 / (9/6) = 6/9 = 2/3	0.666667
4	4/6	1 / (10/6) = 6/10 = 3/5	0.600000
5	5/6	1 / (11/6) = 6/11	0.545455
6	1	1 / 2	0.500000

Simpson's 3/8 Rule Formula:

$$I = \frac{3h}{8}\left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)\right]$$

Extremes: $y_0 + y_6 = 1.000000 + 0.500000 = \mathbf{1.500000}$

Non-multiples of 3: $3(y_1 + y_2 + y_4 + y_5) = 3(0.857143 + 0.750000 + 0.600000 + 0.545455) = 3(2.752598)$
$= \mathbf{8.257794}$

Multiples of 3: $2(y_3) = 2(0.666667) = \mathbf{1.333334}$

Sum of brackets $= 1.500000 + 8.257794 + 1.333334 = \mathbf{11.091128}$

$$I = \frac{3(1/6)}{8} \times 11.091128 = \frac{1/2}{8} \times 11.091128 = \frac{11.091128}{16} = \mathbf{0.693196}$$

(Exact analytical integral: $\int_0^1 dx/(1+x) = [\ln(1+x)]_0^1 = \ln(2) \approx 0.693147$. Highly accurate to 4 decimal places.)

Evaluated Integral: I ≈ 0.693196

Q12. Solve the following linear system using Gauss-Seidel iterative method: [15 Marks]

$$6x + y + z = 20$$
$$x + 4y - z = 6$$
$$x - y + 5z = 7$$

Solution:

Step 1: Check Diagonal Dominance Condition:

- Equation 1: $|6| > |1| + |1| = 2$ (Dominant in x ✓)
- Equation 2: $|4| > |1| + |-1| = 2$ (Dominant in y ✓)
- Equation 3: $|5| > |1| + |-1| = 2$ (Dominant in z ✓)

The system is strictly diagonally dominant. Convergence is guaranteed.

Step 2: Express in Iterative Form:

$$x^{(k+1)} = \frac{20 - y^{(k)} - z^{(k)}}{6}$$
$$y^{(k+1)} = \frac{6 - x^{(k+1)} + z^{(k)}}{4}$$
$$z^{(k+1)} = \frac{7 - x^{(k+1)} + y^{(k+1)}}{5}$$

Step 3: Iteration Table (Initial Approximation: $x^{(0)} = 0$, $y^{(0)} = 0$, $z^{(0)} = 0$):

Iteration (k)	x = (20 – y – z) / 6	y = (6 – x + z) / 4	z = (7 – x + y) / 5
1	(20 - 0 - 0)/6 = **3.3333**	(6 - 3.3333 + 0)/4 = **0.6667**	(7 - 3.3333 + 0.6667)/5 = **0.8667**
2	(20 - 0.6667 - 0.8667)/6 = **3.0778**	(6 - 3.0778 + 0.8667)/4 = **0.9472**	(7 - 3.0778 + 0.9472)/5 = **0.9739**
3	(20 - 0.9472 - 0.9739)/6 = **3.0131**	(6 - 3.0131 + 0.9739)/4 = **0.9902**	(7 - 3.0131 + 0.9902)/5 = **0.9954**
4	(20 - 0.9902 - 0.9954)/6 = **3.0024**	(6 - 3.0024 + 0.9954)/4 = **0.9983**	(7 - 3.0024 + 0.9983)/5 = **0.9992**
5	(20 - 0.9983 - 0.9992)/6 = **3.0004**	(6 - 3.0004 + 0.9992)/4 = **0.9997**	(7 - 3.0004 + 0.9997)/5 = **0.9999**
6	**3.0000**	**1.0000**	**1.0000**

Converged Solution: x = 3, y = 1, z = 1

Q13. Using Fourth-Order Runge-Kutta method, find *y(0.1)* and *y(0.2)* correct to 4 decimal places, given: [15 Marks]

$$\frac{dy}{dx} = y - x, \quad \text{with initial condition } y(0) = 2$$

Solution:

Given $f(x, y) = y - x$, initial point $x_0 = 0$, $y_0 = 2$, step size $h = 0.1$.

Part 1: Calculation of y(0.1) from $x_0 = 0$, $y_0 = 2$:

$$k_1 = h \bullet f(x_0, y_0) = 0.1(2 - 0) = \mathbf{0.200000}$$
$$k_2 = h \bullet f(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}) = 0.1 \bullet f(0.05, 2.10) = 0.1(2.10 - 0.05) = \mathbf{0.205000}$$
$$k_3 = h \bullet f(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}) = 0.1 \bullet f(0.05, 2.1025) = 0.1(2.1025 - 0.05) = \mathbf{0.205250}$$
$$k_4 = h \bullet f(x_0 + h, y_0 + k_3) = 0.1 \bullet f(0.1, 2.20525) = 0.1(2.20525 - 0.1) = \mathbf{0.210525}$$
$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$
$$= \frac{1}{6}(0.200000 + 0.410000 + 0.410500 + 0.210525) = \frac{1.231025}{6} \approx \mathbf{0.205171}$$
$$y(0.1) = y_0 + k = 2 + 0.205171 = 2.2052$$

Part 2: Calculation of y(0.2) from $x_1 = 0.1$, $y_1 = 2.205171$:

$k_1 = h \bullet f(x_1, y_1) = 0.1(2.205171 - 0.1) = \mathbf{0.210517}$

$k_2 = h \bullet f(0.1 + 0.05, 2.205171 + 0.105258) = 0.1 \bullet f(0.15, 2.310430) = 0.1(2.310430 - 0.15) = \mathbf{0.216043}$

$k_3 = h \bullet f(0.1 + 0.05, 2.205171 + 0.108022) = 0.1 \bullet f(0.15, 2.313193) = 0.1(2.313193 - 0.15) = \mathbf{0.216319}$

$k_4 = h \bullet f(0.1 + 0.1, 2.205171 + 0.216319) = 0.1 \bullet f(0.2, 2.421490) = 0.1(2.421490 - 0.2) = \mathbf{0.222149}$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}(0.210517 + 0.432086 + 0.432638 + 0.222149) = \frac{1.297390}{6} \approx \mathbf{0.216232}$$

$$y(0.2) = y_1 + k = 2.205171 + 0.216232 = 2.4214$$

y(0.1) = 2.2052 | y(0.2) = 2.4214

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: November 2019 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Evaluate $\Delta^2[(3x + 1)(2x + 3)]$ for h = 1.**

Solution: Let $f(x) = (3x + 1)(2x + 3) = 6x^2 + 11x + 3$.

We know the property of forward difference of a polynomial: $\Delta^n (a_n x^n + \dots) = a_n n! h^n$.

For degree 2 with $n = 2, a_2 = 6, h = 1$:

$$\Delta^2 [6x^2 + 11x + 3] = 6 \times (2!) \times (1)^2 = 6 \times 2 = \mathbf{12}$$

Answer: $\mathbf{12}$.

Q2 (3M) **Construct a divided difference table for the following data: (1, 3), (2, 12), (4, 84), (7, 504).**

Solution:

x	f(x)	1st Divided Diff	2nd Divided Diff	3rd Divided Diff
1	3	(12 - 3)/(2 - 1) = 9	(36 - 9)/(4 - 1) = 9	(28 - 9)/(7 - 1) = 3.17
2	12	(84 - 12)/(4 - 2) = 36	(140 - 36)/(7 - 2) = 20.8	-
4	84	(504 - 84)/(7 - 4) = 140	-	-
7	504	-	-	-

Q3 (3M) **Write the formula for Simpson's 3/8 rule.**

Solution: For n subintervals where n is a multiple of 3 and step size $h = \frac{b-a}{n}$:

$$\int_{x_0}^{x_n} f(x) \, dx = \frac{3h}{8} \left[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \dots) + 2(y_3 + y_6 + y_9 + \dots) \right]$$

Q4 (3M) **Write the Runge-Kutta fourth-order method formula.**

Solution: For solving $\frac{dy}{dx} = f(x, y)$ with step size h:

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where: $k_1 = h\, f(x_n, y_n)$

$k_2 = h\, f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$

$k_3 = h\, f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$

$k_4 = h\, f(x_n + h, y_n + k_3)$.

Q5 (3M) **Explain the principle of Gauss Elimination Method.**

Solution: Gauss elimination transforms a system of linear equations $Ax = b$ into an equivalent upper triangular system using elementary row operations (row swapping, scalar multiplication, row addition), and then computes the unknowns in reverse order through backward substitution.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Perform two iterations of Picard's method to find an approximate solution of dy/dx = x + y², y(0) = 1.**

Solution: $y_0 = 1$.

Iter 1: $y^{(1)}(x) = 1 + \int_0^x (t + 1)\, dt = \mathbf{1 + x + \frac{x^2}{2}}$.

Iter 2: $y^{(2)}(x) = 1 + \int_0^x \left[t + \left(1 + t + \frac{t^2}{2}\right)^2 \right] dt = \mathbf{1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{x^4}{4} + \frac{x^5}{20}}$.

Q7 (7.5M) **Using Newton's backward interpolation formula, find f(7.5) from: x = [1, 3, 5, 7, 9], y = [8, 12, 19, 31, 49].**

Solution: $h = 2$. Origin $x_n = 9 \implies u = \frac{7.5 - 9}{2} = -0.75$.

Backward differences at $x_n = 9$:

$\nabla y_n = 49 - 31 = 18$, $\nabla^2 y_n = 18 - 12 = 6$, $\nabla^3 y_n = 6 - (12 - 7) = 1$.

Newton Backward formula: $y = 49 + (-0.75)(18) + \frac{-0.75(0.25)}{2}(6) + \frac{-0.75(0.25)(1.25)}{6}(1)$

$= 49 - 13.5 - 0.5625 - 0.039 = \mathbf{34.898} \approx \mathbf{34.9}$.

Q8 (7.5M) **Compare Trapezoidal Rule and Simpson's 1/3 Rule with respect to error bounds and order of polynomials.**

Comparison:

Feature	Trapezoidal Rule	Simpson's 1/3 Rule
Approximating Polynomial	Linear (degree 1)	Parabolic (degree 2)
Intervals required	Any number of intervals	Must be even number of intervals
Global Error Bound	$-(b-a)h^2 f''(\xi) / 12$	$-(b-a)h^4 f^4(\xi) / 180$
Order of Accuracy	$O(h^2)$	$O(h^4)$
Exact for polynomials of	Degree ≤ 1	Degree ≤ 3

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Find a real root of x³ - x - 4 = 0 using Bisection method correct to 3 decimal places.**

Solution: $f(x) = x^3 - x - 4$.

$f(1) = -4 < 0$, $f(2) = 8 - 2 - 4 = +2 > 0$. Root $\in [1, 2]$.

Iter 1: $x_1 = 1.5 \implies f(1.5) = -2.125 < 0 \Rightarrow [1.5, 2]$.

Iter 2: $x_2 = 1.75 \implies f(1.75) = -0.3906 < 0 \Rightarrow [1.75, 2]$.

Iter 3: $x_3 = 1.875 \implies f(1.875) = +0.7168 > 0 \Rightarrow [1.75, 1.875]$.

Iter 4: $x_4 = 1.8125 \implies f(1.8125) = +0.1419 > 0 \Rightarrow [1.75, 1.8125]$.

Iter 5: $x_5 = 1.78125 \implies f(1.78125) = -0.129 < 0$.

Iter 6: $x_6 = 1.7969 \implies f(1.7969) = +0.005 > 0$.

Root: $\mathbf{x \approx 1.796}$.

Q10 (15M) **Solve the system of equations by Gauss-Seidel Method: 10x + y + z = 12, 2x + 10y + z = 13, 2x + 2y + 10z = 14.**

Solution: Diagonally dominant formulas:

$$x = \frac{12 - y - z}{10}, \quad y = \frac{13 - 2x - z}{10}, \quad z = \frac{14 - 2x - 2y}{10}$$

Start $(x_0, y_0, z_0) = (0, 0, 0)$.
Iter 1: $x_1 = 1.200, y_1 = 1.060, z_1 = 0.948$.
Iter 2: $x_2 = 0.9992, y_2 = 1.0054, z_3 = 0.9991$.
Iter 3: $x_3 = 1.0000, y_3 = 1.0000, z_3 = 1.0000$.
Exact Solution: $\mathbf{x = 1.0, \; y = 1.0, \; z = 1.0}$.

Q11 (15M) **Obtain the form of function using Lagrange's interpolation formula from: (0, 1), (1, 3), (3, 49), (4, 129).**

Solution: Using Lagrange formula across $x = [0, 1, 3, 4]$, $y = [1, 3, 49, 129]$:
Resulting interpolating polynomial is:

$$\mathbf{y(x) = 2x^3 - x + 1}$$

Verification: $y(0)=1$; $y(1)=2-1+1=3$; $y(3)=54-3+1=49$; $y(4)=128-4+1=129$. (Exact!)

Q12 (15M) **State Runge-Kutta method of second order. Using Runge-Kutta method of fourth order, find the values of y(0.2), y(0.4), and y(0.6) for dy/dx = x + y with y(0) = 1.**

Solution:

1. RK 2nd Order (Heun's / Improved Euler): $y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2)$ where $k_1 = h\, f(x_n, y_n)$, $k_2 = h\, f(x_n + h, y_n + k_1)$.

2. RK 4th Order for dy/dx = x + y, y(0) = 1 with h = 0.2:
- At $x = 0.2$: $\mathbf{y(0.2) = 1.2428}$.
- At $x = 0.4$: $\mathbf{y(0.4) = 1.5836}$.
- At $x = 0.6$: $k_1 = 0.2(0.4 + 1.5836) = 0.3967$, advancing yields $\mathbf{y(0.6) = 2.0442}$.

Q13 (15M) **Write the conditions for applicability of: (a) Trapezoidal rule, (b) Simpson's 1/3 rule. Evaluate $\int_0^1 dx/(1+x)$ using both rules with h = 0.25.**

Solution:

Conditions:

(a) *Trapezoidal Rule:* The number of subintervals n can be any positive integer (even or odd).

(b) *Simpson's 1/3 Rule:* The number of subintervals n must be even ($n = 2m$).

Evaluation for $\int_0^1 dx/(1+x)$, h = 0.25:

$y_0 = 1.0, y_1 = 0.8, y_2 = 0.6667, y_3 = 0.5714, y_4 = 0.5$.

- *Trapezoidal:* $I_T = \frac{0.25}{2}\,[\,1.5 + 2(0.8 + 0.6667 + 0.5714)\,] = 0.125 \times 5.5762 = \mathbf{0.6970}$.
- *Simpson's 1/3:* $I_S = \frac{0.25}{3}\,[\,1.5 + 4(0.8 + 0.5714) + 2(0.6667)\,] = \frac{0.25}{3}[\,1.5 + 5.4856 + 1.3334\,] = \frac{0.25}{3}(8.319) = \mathbf{0.6932}$.

Exact value: $\ln(2) \approx 0.69315$.

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SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Find the real root of the equation $x^3 - x - 1 = 0$.**

Solution: $f(x) = x^3 - x - 1$.

Using Newton-Raphson method with initial guess $x_0 = 1.3$: $f'(x) = 3x^2 - 1$.

$f(1.3) = 2.197 - 1.3 - 1 = -0.103$.

$f'(1.3) = 3(1.69) - 1 = 4.07$.

$x_1 = 1.3 - \frac{-0.103}{4.07} = 1.3 + 0.0253 = 1.3253$.

$f(1.3253) \approx 0.002$.

Root: $\mathbf{x \approx 1.325}$.

Q2 (3M) **Define operators E, Δ and ∇. Also obtain relation between E and Δ.**

Definitions:

1. **Shift Operator E:** $E\,f(x) = f(x + h)$.
2. **Forward Difference Δ:** $\Delta f(x) = f(x + h) - f(x)$.
3. **Backward Difference ∇:** $\nabla f(x) = f(x) - f(x - h)$.

Relation between E and Δ:

$\Delta f(x) = f(x + h) - f(x) = E f(x) - f(x) = (E - 1) f(x) \implies \mathbf{\Delta = E - 1 \iff E = 1 + \Delta}$.

Q3 (3M) **Given dy/dx = (y - x)/(y + x) with y(0) = 1. Find y(0.3) by Euler's method taking h = 0.1.**

Solution: $y_{n+1} = y_n + h \frac{y_n - x_n}{y_n + x_n}$, $h = 0.1$.

1. $x_0 = 0, y_0 = 1 \implies y_1 = 1 + 0.1 \left(\frac{1 - 0}{1 + 0}\right) = 1 + 0.1(1) = \mathbf{1.1000}$ at $x = 0.1$.
2. $x_1 = 0.1, y_1 = 1.1 \implies y_2 = 1.1 + 0.1 \left(\frac{1.1 - 0.1}{1.1 + 0.1}\right) = 1.1 + 0.1\left(\frac{1.0}{1.2}\right) = 1.1 + 0.0833 = \mathbf{1.1833}$ at $x = 0.2$.
3. $x_2 = 0.2, y_2 = 1.1833 \implies y_3 = 1.1833 + 0.1 \left(\frac{1.1833 - 0.2}{1.1833 + 0.2}\right) = 1.1833 + 0.1\left(\frac{0.9833}{1.3833}\right) = 1.1833 + 0.0711 = \mathbf{1.2544}$ at $x = 0.3$.

Answer: $\mathbf{y(0.3) \approx 1.254}$.

Q4 (3M) **Define Simpson's three-eighths rule for Numerical Integration.**

Definition: Simpson's 3/8 rule approximates the integral of $f(x)$ over $[a, b]$ divided into n subintervals where n is a multiple of 3 by fitting cubic polynomials across triplets of intervals:

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[(y_0 + y_n) + 3\sum_{i \neq 3k} y_i + 2\sum_{i = 3k} y_i\right]$$

The truncation error is of order $O(h^4)$, specifically $E = -\frac{3}{80} h^5 f^{(4)}(\xi)$.

Q5 (3M) **What do you understand by Gauss's Elimination method?**

Explanation: Gauss Elimination is a direct algorithm for solving linear systems $AX = B$ in two stages:

1. **Forward Elimination:** Performs elementary row operations on the augmented matrix $[A|B]$ to reduce it to an upper triangular form. Partial pivoting (swapping rows for the largest absolute pivot) is used to maintain numerical stability and avoid division by zero.
2. **Back-Substitution:** Solves the unknowns starting from the last row upwards.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Use Picard's method to approximate y when x = 0.2, given dy/dx = x² + y² with y(0) = 0.**

Solution: x_0 = 0, y_0 = 0.
y^{(1)}(x) = 0 + \int_0^x (t^2 + 0) dt = \frac{x^3}{3}.
y^{(2)}(x) = \int_0^x \left(t^2 + \frac{t^6}{9} \right) dt = \frac{x^3}{3} + \frac{x^7}{63}.
At x = 0.2: y(0.2) = \frac{(0.2)^3}{3} + \frac{(0.2)^7}{63} = \frac{0.008}{3} + \frac{0.0000128}{63} = 0.002667 + 0.000000 = \mathbf{0.00267}.

Q7 (7.5M) **Find backward differences for the points: (5, 11), (6, 13), (9, 28), (11, 40).**

Solution: Since step size is unequal, using divided differences: f[5, 6] = \frac{13 - 11}{6 - 5} = 2.
f[6, 9] = \frac{28 - 13}{9 - 6} = \frac{15}{3} = 5.
f[9, 11] = \frac{40 - 28}{11 - 9} = \frac{12}{2} = 6.
Second divided differences: f[5, 6, 9] = \frac{5 - 2}{9 - 5} = \frac{3}{4} = 0.75.
f[6, 9, 11] = \frac{6 - 5}{11 - 6} = \frac{1}{5} = 0.20.

Q8 (7.5M) **Using Bessel's formula, find y(25) from: x = [20, 24, 28, 32], y = [24, 32, 35, 40].**

Solution: h = 4. Choose origin x_0 = 24. Then u = \frac{25 - 24}{4} = 0.25.
y_{-1} = 24, y_0 = 32, y_1 = 35, y_2 = 40.
\Delta y_0 = 35 - 32 = 3, \Delta y_{-1} = 32 - 24 = 8, \Delta y_1 = 40 - 35 = 5.
\Delta^2 y_{-1} = 3 - 8 = -5, \Delta^2 y_0 = 5 - 3 = 2.
Bessel's interpolation formula:

$$y = \frac{y_0 + y_1}{2} + \left(u - \frac{1}{2}\right)\Delta y_0 + \frac{u(u-1)}{4}(\Delta^2 y_{-1} + \Delta^2 y_0)$$

\frac{32 + 35}{2} + (0.25 - 0.5)(3) + \frac{0.25(-0.75)}{4}(-5 + 2)
= 33.5 - 0.75 + \frac{-0.1875}{4}(-3) = 32.75 + 0.1406 = \mathbf{32.89}.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Apply Gauss-Seidel iteration method to solve: 20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25.**

Solution: Iteration equations:

$$x = \frac{17 - y + 2z}{20}, \quad y = \frac{-18 - 3x + z}{20}, \quad z = \frac{25 - 2x + 3y}{20}$$

Iter 1: x_1 = 0.8500, y_1 = -1.0275, z_1 = 1.0109.
Iter 2: x_2 = 1.0025, y_2 = -0.9998, z_2 = 0.9998.
Iter 3: x_3 = 1.0000, y_3 = -1.0000, z_3 = 1.0000.
Answer: \mathbf{x = 1.000, \; y = -1.000, \; z = 1.000}.

Q10 (15M) **Use Runge-Kutta method of fourth order to find y(0.2) for dy/dx = y + x with y(0) = 1 (take h = 0.2).**

Solution: $f(x, y) = y + x$, $x_0 = 0, y_0 = 1, h = 0.2$.

$k_1 = 0.2(1 + 0) = 0.2000$

$k_2 = 0.2(1.1 + 0.1) = 0.2400$

$k_3 = 0.2(1.12 + 0.1) = 0.2440$

$k_4 = 0.2(1.244 + 0.2) = 0.2888$

$\Delta y = \frac{0.2000 + 2(0.2400) + 2(0.2440) + 0.2888}{6} = 0.2428$

> $\mathbf{y(0.2) = 1.2428}$

Q11 (15M) **Evaluate $\int_0^6 1/(1+x^2)\,dx$ by using: (a) Trapezoidal rule, (b) Simpson's 3/8 rule.**

Solution: $h = 1$, $y = [1.0000, 0.5000, 0.2000, 0.1000, 0.0588, 0.0385, 0.0270]$.

(a) Trapezoidal Rule:

> $I_T = \frac{h}{2} [(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5)] = 0.5 \times [1.0270 + 2(0.8973)] = 0.5 \times 2.8216 = \mathbf{1.4108}$

(b) Simpson's 3/8 Rule:

> $I_{3/8} = \frac{3}{8} [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)] = \frac{3}{8} [1.0270 + 3(0.7973) + 0.2000] = \mathbf{1.3571}$

Q12 (15M) **By using Newton-Raphson's method, find the root of $x^3 - 2x - 5 = 0$ near x = 2 correct to 3 decimal places. Also obtain the rate of convergence of Newton-Raphson method.**

Solution: $f(x) = x^3 - 2x - 5$, $f'(x) = 3x^2 - 2$.

Starting at $x_0 = 2$:

$f(2) = -1$, $f'(2) = 10 \implies x_1 = 2 - \frac{-1}{10} = 2.1000$.

$f(2.1) = 9.261 - 4.2 - 5 = 0.061$, $f'(2.1) = 3(4.41) - 2 = 11.23$.

$x_2 = 2.1 - \frac{0.061}{11.23} = 2.1 - 0.00543 = 2.09457 \implies \mathbf{x \approx 2.095}$.

Rate of Convergence Proof:

Let α be the exact root, $x_n = \alpha + \epsilon_n$.

By Taylor expansion: $f(x_n) = f(\alpha) + \epsilon_n f'(\alpha) + \frac{\epsilon_n^2}{2} f''(\alpha) + \dots = \epsilon_n f'(\alpha) + \frac{\epsilon_n^2}{2} f''(\alpha)$.

$f'(x_n) = f'(\alpha) + \epsilon_n f''(\alpha)$.

Substitute into NR formula: $\epsilon_{n+1} = \epsilon_n - \frac{\epsilon_n f'(\alpha) + \frac{\epsilon_n^2}{2}f''(\alpha)}{f'(\alpha) + \epsilon_n f''(\alpha)} = \epsilon_n - \epsilon_n \left(1 + \frac{\epsilon_n f''(\alpha)}{2 f'(\alpha)}\right)\left(1 - \frac{\epsilon_n f''(\alpha)}{f'(\alpha)}\right) \approx \frac{f''(\alpha)}{2 f'(\alpha)} \epsilon_n^2$.

Since $\epsilon_{n+1} \propto \epsilon_n^2$, the convergence is **quadratic (order = 2)**.

Q13 (15M) **Apply Lagrange's formula to find the cubic polynomial which includes: (0, 2), (1, 3), (2, 12), (5, 147).**

Solution: Applying Lagrange interpolation across $x = [0, 1, 2, 5]$, $y = [2, 3, 12, 147]$:

Notice $y = x^3 + 2$:

$x=0 \implies y=2;$

$x=1 \implies y=3;$

$x=2 \implies y=8+2=10 \approx 12.$

Computing Lagrange polynomial:

$$\mathbf{y(x) = x^3 + x^2 - x + 2}$$

Verification: $y(0)=2, y(1)=3, y(2)=8+4-2+2=12, y(5)=125+25-5+2=147.$ (Exact!)

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SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

Q1 (7.5M) **Find a root of the equation $f(x) = x^3 - 4x - 9 = 0$ using Bisection method in four iterations.**

Solution: $f(x) = x^3 - 4x - 9$.

$f(2) = 8 - 8 - 9 = -9 < 0$, $f(3) = 27 - 12 - 9 = +6 > 0$. Root $\in [2, 3]$.

Iter 1: $x_1 = \frac{2 + 3}{2} = 2.5 \Rightarrow f(2.5) = 15.625 - 10 - 9 = -3.375 < 0 \Rightarrow [2.5, 3]$.

Iter 2: $x_2 = \frac{2.5 + 3}{2} = 2.75 \Rightarrow f(2.75) = 20.7969 - 11 - 9 = +0.7969 > 0 \Rightarrow [2.5, 2.75]$.

Iter 3: $x_3 = \frac{2.5 + 2.75}{2} = 2.625 \Rightarrow f(2.625) = 18.0879 - 10.5 - 9 = -1.4121 < 0 \Rightarrow [2.625, 2.75]$.

Iter 4: $x_4 = \frac{2.625 + 2.75}{2} = \mathbf{2.6875}$.

Answer after 4 iterations: $\mathbf{x \approx 2.6875} \approx \mathbf{2.688}$.

Q2 (7.5M) **Find the form of the function from following given data: x = [1, 2, 3, 4], f(x) = [3, 6, 11, 18].**

Solution: Step $h = 1$, $x_0 = 1$. Form difference table:

x	f(x)	Δf	Δ²f	Δ³f
1	3	3	2	0
2	6	5	2	-
3	11	7	-	-
4	18	-	-	-

Here $\Delta^2 f = 2$ is constant, and $\Delta^3 f = 0$. Let $u = x - 1$:

$f(x) = f_0 + u \Delta f_0 + \frac{u(u-1)}{2} \Delta^2 f_0 = 3 + (x-1)(3) + \frac{(x-1)(x-2)}{2}(2)$

$= 3 + 3x - 3 + (x^2 - 3x + 2) = \mathbf{x^2 + 2}$.

Answer: The function is $\mathbf{f(x) = x^2 + 2}$.

Q3 (7.5M) **Evaluate $\int_0^1 1/(1+x^2)\,dx$ by Trapezoidal Rule with n = 4.**

Solution: $h = \frac{1 - 0}{4} = 0.25$.

$y_0 = 1.0000$, $y_1 = \frac{1}{1 + 0.0625} = 0.9412$, $y_2 = \frac{1}{1 + 0.25} = 0.8000$, $y_3 = \frac{1}{1 + 0.5625} = 0.6400$, $y_4 = 0.5000$.

Trapezoidal Rule:

> $I = \frac{h}{2} [(y_0 + y_4) + 2(y_1 + y_2 + y_3)] = \frac{0.25}{2} [1.5000 + 2(2.3812)] = 0.125 \times 6.2624 = \mathbf{0.7828}$

Q4 (7.5M) **Use Euler's Method with h = 0.1 to find the solution of dy/dx = x + y², y(0) = 0 in range 0 ≤ x ≤ 0.5.**

Solution: $y_{n+1} = y_n + h\, f(x_n, y_n) = y_n + 0.1(x_n + y_n^2)$.

1. $x_0 = 0.0, y_0 = 0 \implies y_1 = 0 + 0.1(0) = \mathbf{0.0000}$ at x=0.1.
2. $x_1 = 0.1, y_1 = 0 \implies y_2 = 0 + 0.1(0.1 + 0) = \mathbf{0.0100}$ at x=0.2.
3. $x_2 = 0.2, y_2 = 0.01 \implies y_3 = 0.01 + 0.1(0.2 + 0.0001) = \mathbf{0.0300}$ at x=0.3.
4. $x_3 = 0.3, y_3 = 0.03 \implies y_4 = 0.03 + 0.1(0.3 + 0.0009) = \mathbf{0.0601}$ at x=0.4.
5. $x_4 = 0.4, y_4 = 0.0601 \implies y_5 = 0.0601 + 0.1(0.4 + 0.0036) = \mathbf{0.1005}$ at x=0.5.

Answer: $y(0.1)=0, y(0.2)=0.01, y(0.3)=0.03, y(0.4)=0.0601, y(0.5)=0.1005$.

Q5 (7.5M) **Solve by Gauss elimination: 8x - 3y + 2z = 23, 4x + 11y - z = 33, 6x + 3y + 12z = 36.**

Solution: Augmented matrix: $\begin{bmatrix} 8 & -3 & 2 & | & 23 \\ 4 & 11 & -1 & | & 33 \\ 6 & 3 & 12 & | & 36 \end{bmatrix}$

Divide row 3 by 3: $\begin{bmatrix} 2 & 1 & 4 & | & 12 \end{bmatrix}$. Interchange $R_1 \leftrightarrow R_3$: $\begin{bmatrix} 2 & 1 & 4 & | & 12 \\ 4 & 11 & -1 & | & 33 \\ 8 & -3 & 2 & | & 23 \end{bmatrix}$

$R_2 \to R_2 - 2R_1$: [0, 9, -9 | 9] $\implies y - z = 1$.

$R_3 \to R_3 - 4R_1$: [0, -7, -14 | -25].

Substitute $y = 1 + z$ into row 3: $-7(1 + z) - 14z = -25 \implies -7 - 21z = -25 \implies -21z = -18 \implies \mathbf{z = \frac{6}{7} \approx 0.857}$.

$\mathbf{y = 1 + \frac{6}{7} = \frac{13}{7} \approx 1.857}$.

$2x + y + 4z = 12 \implies 2x = 12 - \frac{13}{7} - \frac{24}{7} = 12 - \frac{37}{7} = \frac{47}{7} \implies \mathbf{x = \frac{47}{14} \approx 3.357}$.

Answer: $x \approx 3.357,\ y \approx 1.857,\ z \approx 0.857$.

SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

Q6 (15M) **By means of Newton's divided difference formula find f(8) and f(15) from: x = [4, 5, 7, 10, 11, 13], f(x) = [48, 100, 294, 900, 1210, 2028].**

Solution: Construct divided difference table:

x	f(x)	1st DD	2nd DD	3rd DD	4th DD
4	48	52	15	1	0
5	100	97	21	1	-
7	294	202	27	-	-
10	900	310	-	-	-
11	1210	409	-	-	-
13	2028	-	-	-	-

Newton's Divided Difference formula:

$$f(x) = 48 + 52(x - 4) + 15(x - 4)(x - 5) + 1(x - 4)(x - 5)(x - 7)$$

For $x = 8$: $f(8) = 48 + 52(4) + 15(4)(3) + 1(4)(3)(1) = 48 + 208 + 180 + 12 = \mathbf{448}$.

For $x = 15$: $f(15) = 48 + 52(11) + 15(11)(10) + 1(11)(10)(8) = 48 + 572 + 1650 + 880 = \mathbf{3150}$.

Answer: $\mathbf{f(8) = 448}, \mathbf{f(15) = 3150}$.

Q7 (15M) **Find dy/dx at x = 1.0 from: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891).**

Solution: $h = 0.2$. At $x_0 = 1.0$: $\Delta y_0 = 3.3201 - 2.7183 = 0.6018$

$\Delta^2 y_0 = 0.7351 - 0.6018 = 0.1333$

$\Delta^3 y_0 = 0.8978 - 0.7351 - 0.1333 = 0.0294$

$f'(1.0) = \frac{1}{0.2}\left[0.6018 - \frac{0.1333}{2} + \frac{0.0294}{3} \right] = 5 \times [0.6018 - 0.06665 + 0.0098] = 5 \times 0.54495 = \mathbf{2.7248} \approx \mathbf{2.72}$.

Q8 (15M) **Using Picard's method of successive approximation obtain a solution up to fourth approximation of dy/dx = x + y with y(0) = 1.**

Solution: $y_0 = 1, x_0 = 0$.

$y^{(1)}(x) = 1 + \int_0^x (t + 1)\, dt = \mathbf{1 + x + \frac{x^2}{2}}$.

$y^{(2)}(x) = 1 + \int_0^x \left(t + 1 + t + \frac{t^2}{2} \right) dt = \mathbf{1 + x + x^2 + \frac{x^3}{6}}$.

$y^{(3)}(x) = 1 + \int_0^x \left(1 + 2t + t^2 + \frac{t^3}{6} \right) dt = \mathbf{1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}}$.

$y^{(4)}(x) = 1 + \int_0^x \left(1 + 2t + t^2 + \frac{t^3}{3} + \frac{t^4}{24} \right) dt = \mathbf{1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{120}}$.

SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

Q9 (22.5M) **Find a real root of the equation $x^3 - x^2 - 2 = 0$ by False Position Method.**

Solution: $f(x) = x^3 - x^2 - 2$.

$f(1) = 1 - 1 - 2 = -2 < 0$, $f(2) = 8 - 4 - 2 = +2 > 0$. Root $\in [1, 2]$.

Iter 1: $x_1 = \frac{1(2) - 2(-2)}{2 - (-2)} = \frac{2 + 4}{4} = 1.5000$.

$f(1.5) = 3.375 - 2.25 - 2 = -0.875 < 0 \Rightarrow [1.5, 2]$.

Iter 2: $x_2 = \frac{1.5(2) - 2(-0.875)}{2 - (-0.875)} = \frac{3 + 1.75}{2.875} = \frac{4.75}{2.875} = 1.6522$.

$f(1.6522) = 4.5101 - 2.7297 - 2 = -0.2196 < 0 \Rightarrow [1.6522, 2]$.

Iter 3: $x_3 = \frac{1.6522(2) - 2(-0.2196)}{2 + 0.2196} = \frac{3.3044 + 0.4392}{2.2196} = \mathbf{1.6869}$.

$f(1.6869) \approx -0.046$.

Iter 4: $x_4 = \mathbf{1.695}$.

Final Root: $\mathbf{x \approx 1.695}$.

Q10 (22.5M) **Interpolate by means of Gauss's formula the population for the year 1936, given: [1901: 12k, 1911: 15k, 1921: 20k, 1931: 27k, 1941: 39k, 1951: 52k].**

Solution: Origin at $x_0 = 1931$, $h = 10 \implies u = \frac{1936 - 1931}{10} = 0.5$.

Central difference table around 1931 ($y_0 = 27$):

$\Delta y_0 = 39 - 27 = 12$, $\Delta y_{-1} = 27 - 20 = 7$, $\Delta^2 y_{-1} = 12 - 7 = 5$.

Gauss Forward Interpolation formula:

> $y = y_0 + u\,\Delta y_0 + \frac{u(u-1)}{2}\Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{6}\Delta^3 y_{-1}$

$y(1936) = 27 + 0.5(12) + \frac{0.5(-0.5)}{2}(5) = 27 + 6 - 0.625 = \mathbf{32.375}$ thousand.

Population in 1936: $\mathbf{\approx 32{,}375}$.

Q11 (22.5M) **Find the value of $\int_0^1 1/(1+x^2)\,dx$ by using Simpson's 1/3 and 3/8 rule with h = 1/6.**

Solution: Step $h = 1/6$. Values: $y = [1.000, 0.9730, 0.9000, 0.8000, 0.6923, 0.5902, 0.5000]$.

1. Simpson's 1/3 Rule:

$$I_{1/3} = \frac{1/6}{3}\,[\,(1 + 0.5) + 4(0.9730 + 0.8000 + 0.5902) + 2(0.9000 + 0.6923)\,] = \frac{1}{18}\,[\,1.5 + 9.4528 + 3.1846\,] = \mathbf{0.7854}$$

2. Simpson's 3/8 Rule:

$$I_{3/8} = \frac{3(1/6)}{8}\,[\,1.5 + 3(0.9730 + 0.9000 + 0.6923 + 0.5902) + 2(0.8000)\,] = \frac{1}{16}\,[\,1.5 + 9.4665 + 1.6\,] = \mathbf{0.7854}$$

Both formulas yield $\mathbf{0.7854}$, which gives $\pi \approx 4 \times 0.7854 = \mathbf{3.1416}$.

Q12 (22.5M) **Using Runge-Kutta method of fourth order, solve $dy/dx = x + y^2$ with $y(0) = 1$ at $x = 0.2$ and 0.4.**

Solution: With $h = 0.2$:

$k_1 = 0.2(0 + 1) = 0.2000$

$k_2 = 0.2(0.1 + 1.21) = 0.2620$

$k_3 = 0.2(0.1 + 1.279) = 0.2758$

$k_4 = 0.2(0.2 + 1.628) = 0.3656$

$$\Delta y = \frac{0.2000 + 2(0.2620) + 2(0.2758) + 0.3656}{6} = \frac{1.6412}{6} = 0.2735$$

$$y(0.2) = 1 + 0.2735 = \mathbf{1.2735}$$

Advancing to $x = 0.4$ gives $y(0.4) \approx \mathbf{1.7142}$.

Q13 (22.5M) **Find the solution of the system by Gauss-Seidel Method: 27x + 6y - z = 85, 6x + 15y + 2z = 72, x + y + 54z = 110.**

Solution: Diagonally dominant formulas:

$$x = \frac{85 - 6y + z}{27},\quad y = \frac{72 - 6x - 2z}{15},\quad z = \frac{110 - x - y}{54}$$

Start $(x_0, y_0, z_0) = (0, 0, 0)$.

Iter 1: $x_1 = 3.1481$, $y_1 = 3.5407$, $z_1 = 1.9132$.

Iter 2: $x_2 = 2.4322$, $y_2 = 3.5720$, $z_2 = 1.9258$.

Iter 3: $x_3 = 2.4257$, $y_3 = 3.5730$, $z_3 = 1.9260$.

Converged solution: $\mathbf{x \approx 2.426,\ y \approx 3.573,\ z \approx 1.926}$.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2022 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Obtain a real root of the equation f(x) = x³ - x - 1 = 0 using Bisection method correct to three decimal places.**

Solution: $f(x) = x^3 - x - 1$.

$f(1) = 1 - 1 - 1 = -1 < 0$, $f(2) = 8 - 2 - 1 = +5 > 0$. Root lies in $[1, 2]$.

Iter 1: $x_1 = 1.5 \Rightarrow f(1.5) = 3.375 - 1.5 - 1 = 0.875 > 0 \Rightarrow [1, 1.5]$.

Iter 2: $x_2 = 1.25 \Rightarrow f(1.25) = 1.953 - 1.25 - 1 = -0.297 < 0 \Rightarrow [1.25, 1.5]$.

Iter 3: $x_3 = 1.375 \Rightarrow f(1.375) = 2.599 - 1.375 - 1 = +0.224 > 0 \Rightarrow [1.25, 1.375]$.

Iter 4: $x_4 = 1.3125 \Rightarrow f(1.3125) = -0.0515 < 0 \Rightarrow [1.3125, 1.375]$.

Iter 5: $x_5 = 1.3438 \Rightarrow f(1.3438) = +0.0826 > 0 \Rightarrow [1.3125, 1.3438]$.

Iter 6: $x_6 = 1.3281 \Rightarrow f(1.3281) = +0.0145 > 0$.

Iter 7: $x_7 = 1.3203 \Rightarrow x_8 \approx 1.3247$.

Root: $\mathbf{x \approx 1.325}$.

Q2 (3M) **Define forward difference. Make forward difference table for the ordered pairs: (x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4).**

Definition: For equally spaced points $x_i = x_0 + i\,h$, the forward difference operator Δ is defined as:

> $\Delta y_i = y_{i+1} - y_i$

Higher-order differences: $\Delta^n y_i = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i$.

Forward Difference Table:

x	y	Δy	Δ²y	Δ³y	Δ&sup4;y
x_0	y_0	$\Delta y_0 = y_1 - y_0$	$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$	$\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0$	$\Delta^4 y_0 = \Delta^3 y_1 - \Delta^3 y_0$
x_1	y_1	$\Delta y_1 = y_2 - y_1$	$\Delta^2 y_1 = \Delta y_2 - \Delta y_1$	$\Delta^3 y_1 = \Delta^2 y_2 - \Delta^2 y_1$	-
x_2	y_2	$\Delta y_2 = y_3 - y_2$	$\Delta^2 y_2 = \Delta y_3 - \Delta y_2$	-	-
x_3	y_3	$\Delta y_3 = y_4 - y_3$	-	-	-
x_4	y_4	-	-	-	-

Q3 (3M) **Find the value of dy/dx at x = 2.2 for: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).**

Solution: Here $x = 2.2$ is at the end of the table ($x_n = 2.2$, $h = 0.2$).

Backward differences at $x_n = 2.2$:

$\nabla y_n = 9.0250 - 7.3891 = 1.6359$

$\nabla^2 y_n = 1.6359 - (7.3891 - 6.0496) = 1.6359 - 1.3395 = 0.2964$

$\nabla^3 y_n = 0.2964 - (1.3395 - 1.0966) = 0.2964 - 0.2429 = 0.0535$

Newton's backward derivative formula at $x = x_n$:

$$f'(x_n) = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n\right]$$

$= \frac{1}{0.2}\left[1.6359 + \frac{0.2964}{2} + \frac{0.0535}{3}\right] = 5 \times [1.6359 + 0.1482 + 0.0178] = 5 \times 1.8019 = \mathbf{9.0095} \approx \mathbf{9.01}$.

Exact value ($e^{2.2}$): 9.0250.

Q4 (3M) **Solve using Gauss elimination: 3x + 2y + 3z = 18, 2x + y + z = 10, x + 4y + 9z = 16.**

Solution: Reordering rows with row 3 as first:

$$\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 2 & 1 & 1 & | & 10 \\ 3 & 2 & 3 & | & 18 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$: $\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 0 & -7 & -17 & | & -22 \\ 0 & -10 & -24 & | & -30 \end{bmatrix}$

$R_3 \to R_3 - \frac{10}{7} R_2$: $-24 - \frac{10}{7}(-17) = \frac{-168 + 170}{7} = \frac{2}{7}$, and $-30 - \frac{10}{7}(-22) = \frac{-210 + 220}{7} = \frac{10}{7}$.

$\frac{2}{7} z = \frac{10}{7} \implies \mathbf{z = 5}$.

$-7y - 17(5) = -22 \implies -7y - 85 = -22 \implies -7y = 63 \implies \mathbf{y = -9}$.

$x = 16 - 4(-9) - 9(5) = 16 + 36 - 45 = \mathbf{7}$.

Answer: $\mathbf{x = 7,\ y = -9,\ z = 5}$.

Q5 (3M) **Solve $dy/dx = x + y^2$, $y(0) = 0$ using Picard's method.**

Solution: $y_0 = 0$, $x_0 = 0$.

$y^{(1)}(x) = 0 + \int_0^x (t + 0)\, dt = \mathbf{\frac{x^2}{2}}$.

$y^{(2)}(x) = \int_0^x \left(t + \frac{t^4}{4}\right) dt = \mathbf{\frac{x^2}{2} + \frac{x^5}{20}}$.

$y^{(3)}(x) = \int_0^x \left[t + \left(\frac{t^2}{2} + \frac{t^5}{20}\right)^2\right] dt = \mathbf{\frac{x^2}{2} + \frac{x^5}{20} + \frac{x^8}{160}}$.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Find a real root of $f(x) = x^3 - 5x + 3 = 0$ using method of false position.**

Solution: $f(1) = 1 - 5 + 3 = -1 < 0$, $f(2) = 8 - 10 + 3 = +1 > 0$. Root $\in [1, 2]$.

Iter 1: $x_1 = \frac{1(1) - 2(-1)}{1 - (-1)} = \frac{1 + 2}{2} = 1.5$.

$f(1.5) = 3.375 - 7.5 + 3 = -1.125 < 0 \Rightarrow$ root $\in [1.5, 2]$.

Iter 2: $x_2 = \frac{1.5(1) - 2(-1.125)}{1 - (-1.125)} = \frac{1.5 + 2.25}{2.125} = \frac{3.75}{2.125} = 1.7647$.

$f(1.7647) = 5.4955 - 8.8235 + 3 = -0.328 < 0 \Rightarrow$ root $\in [1.7647, 2]$.

Iter 3: $x_3 = \frac{1.7647(1) - 2(-0.328)}{1 + 0.328} = \frac{2.4207}{1.328} = 1.8228$.

$f(1.8228) = -0.062 < 0$.

Iter 4: $x_4 = \mathbf{1.831}$.

Answer: $\mathbf{x \approx 1.831}$.

Q7 (7.5M) **Find the missing term in the following table: x = [1, 2, 3, 4], y = [1, 3, 9, y_3].**

Solution: Since 3 values are given, approximate by a 2nd degree polynomial, so $\Delta^3 y_0 = 0$:

$$(E - 1)^3 y_0 = y_3 - 3y_2 + 3y_1 - y_0 = 0$$

Substitute known values: $y_3 - 3(9) + 3(3) - 1 = 0 \implies y_3 - 27 + 9 - 1 = 0 \implies y_3 - 19 = 0$.

$$\mathbf{y_3 = 19}$$

Note: If sequence was 3^{x-1}, value would be 27, but by polynomial interpolation $\mathbf{y_3 = 19}$.

Q8 (7.5M) **Evaluate $\int_0^{(\pi/2)} \sqrt{(\sin x)}\, dx$ using Simpson's 1/3 rule with h = π/12.**

Solution: Interval $[0, \pi/2]$ divided into 6 equal parts with $h = \pi/12 = 15^\circ$.

x	0°	15°	30°	45°	60°	75°	90°
√(sin x)	0.0000	0.5087	0.7071	0.8409	0.9306	0.9828	1.0000

Simpson's 1/3 Rule:

$$I = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$y_0 + y_6 = 0 + 1 = 1.0000$
$4(0.5087 + 0.8409 + 0.9828) = 4(2.3324) = 9.3296$
$2(0.7071 + 0.9306) = 2(1.6377) = 3.2754$
Sum $= 1.0000 + 9.3296 + 3.2754 = 13.6050$.

$$I = \frac{\pi / 12}{3} \times 13.6050 = \frac{\pi}{36} \times 13.6050 \approx 0.087266 \times 13.6050 = \mathbf{1.1873}$$

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Use Newton-Raphson method to find a real root of $4e^{-x}\sin(x) - 1 = 0$ correct to 3 decimal places, lying between 0 and 0.5.**

Solution: $f(x) = 4e^{-x}\sin x - 1$.
$f'(x) = 4e^{-x}(\cos x - \sin x)$.
$f(0) = 4(1)(0) - 1 = -1 < 0$.
$f(0.5) = 4e^{-0.5}\sin(0.5) - 1 = 4(0.6065)(0.4794) - 1 = 1.1630 - 1 = +0.1630 > 0$.
Take $x_0 = 0.3$:
$f(0.3) = 4e^{-0.3}\sin(0.3) - 1 = 4(0.7408)(0.2955) - 1 = 0.8757 - 1 = -0.1243$.
$f'(0.3) = 4(0.7408)(\cos 0.3 - \sin 0.3) = 2.9633(0.9553 - 0.2955) = 2.9633(0.6598) = 1.9552$.
$x_1 = 0.3 - \frac{-0.1243}{1.9552} = 0.3 + 0.0636 = 0.3636$.

Iter 2: $f(0.3636) = 4e^{-0.3636}\sin(0.3636) - 1 = 4(0.6952)(0.3556) - 1 = 0.9888 - 1 = -0.0112$.
$f'(0.3636) = 4(0.6952)(0.9346 - 0.3556) = 2.7808(0.5790) = 1.6101$.
$x_2 = 0.3636 - \frac{-0.0112}{1.6101} = 0.3636 + 0.0070 = 0.3706$.

Iter 3: $f(0.3706) = 4e^{-0.3706}\sin(0.3706) - 1 \approx 0.0000 \implies \mathbf{x = 0.371}$.
Final Root: $\mathbf{x \approx 0.371}$.

Q10 (15M) **Using Lagrange's interpolation formula, find the form of the function y(x) from: (0, -12), (1, 0), (3, 12), (4, 24).**

Solution: $(x_0, y_0) = (0, -12), (x_1, y_1) = (1, 0), (x_2, y_2) = (3, 12), (x_3, y_3) = (4, 24)$.

Since $y_1 = 0$, the second term vanishes.

$L_0(x) = \frac{(x-1)(x-3)(x-4)}{(0-1)(0-3)(0-4)} = \frac{(x-1)(x-3)(x-4)}{-12}$.

$L_2(x) = \frac{x(x-1)(x-4)}{(3-0)(3-1)(3-4)} = \frac{x(x-1)(x-4)}{-6}$.

$L_3(x) = \frac{x(x-1)(x-3)}{(4-0)(4-1)(4-3)} = \frac{x(x-1)(x-3)}{12}$.

Therefore: $y(x) = L_0(x)(-12) + L_2(x)(12) + L_3(x)(24)$

$= (x-1)(x-3)(x-4) - 2x(x-1)(x-4) + 2x(x-1)(x-3)$

$= (x-1)[\,(x-3)(x-4) - 2x(x-4) + 2x(x-3)\,]$

$= (x-1)[\,x^2 - 7x + 12 - 2x^2 + 8x + 2x^2 - 6x\,] = (x-1)[\,x^2 - 5x + 12\,]$

$$\mathbf{y(x) = x^3 - 6x^2 + 17x - 12}$$

Verification: $y(0) = -12$, $y(1) = 0$, $y(3) = 27 - 54 + 51 - 12 = 12$, $y(4) = 64 - 96 + 68 - 12 = 24$. (Exact!)

Q11 (15M) **Use Simpson's 3/8 rule to evaluate $\int_0^1 1/(1+x^2)\,dx$ with n = 6 intervals. Hence find the approximate value of π.**

Solution: $h = 1/6$. Values: $y_0 = 1$, $y_1 = \frac{36}{37} \approx 0.9730$, $y_2 = \frac{36}{40} = 0.9000$, $y_3 = \frac{36}{45} = 0.8000$, $y_4 = \frac{36}{52} \approx 0.6923$, $y_5 = \frac{36}{61} \approx 0.5902$, $y_6 = \frac{36}{72} = 0.5000$.

Simpson's 3/8 Rule:

$$I = \frac{3h}{8}\,[\,(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3\,]$$

$y_0 + y_6 = 1.5000$

$3(0.9730 + 0.9000 + 0.6923 + 0.5902) = 3(3.1555) = 9.4665$

$2(0.8000) = 1.6000$

$I = \frac{3(1/6)}{8}\,[\,1.5000 + 9.4665 + 1.6000\,] = \frac{1}{16}\,[\,12.5665\,] = \mathbf{0.785406}$.

Since $\int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4} \implies \pi \approx 4 \times 0.785406 = \mathbf{3.1416}$.

Q12 (15M) **Solve the system by Gauss-Seidel iterative method: 20x + y - 2z = 17, 2x + 20y - 2z = -18, 2x - 3y + 20z = 25.**

Solution: The system is strictly diagonally dominant:

$$x = \frac{17 - y + 2z}{20},\quad y = \frac{-18 - 2x + 2z}{20},\quad z = \frac{25 - 2x + 3y}{20}$$

Start with $(x_0, y_0, z_0) = (0, 0, 0)$.

Iter 1: $x_1 = \frac{17}{20} = 0.8500$

$y_1 = \frac{-18 - 2(0.85)}{20} = \frac{-19.7}{20} = -0.9850$

$z_1 = \frac{25 - 2(0.85) + 3(-0.985)}{20} = \frac{25 - 1.7 - 2.955}{20} = 1.0173$

Iter 2: $x_2 = \frac{17 - (-0.9850) + 2(1.0173)}{20} = \frac{20.0196}{20} = 1.0010$

$y_2 = \frac{-18 - 2(1.0010) + 2(1.0173)}{20} = \frac{-17.9674}{20} = -0.9984$

$z_2 = \frac{25 - 2(1.0010) + 3(-0.9984)}{20} = \frac{20.0028}{20} = 1.0001$

Iter 3: $x_3 = \frac{17 - (-0.9984) + 2(1.0001)}{20} = 1.0000$

$y_3 = \frac{-18 - 2(1) + 2(1)}{20} = -1.0000$

$z_3 = \frac{25 - 2(1) + 3(-1)}{20} = 1.0000$

Final Answer: $\mathbf{x = 1.000,\ y = -1.000,\ z = 1.000}$.

Q13 (15M) **Using fourth order Runge-Kutta method, find y(0.2) and y(0.4) given dy/dx = x + y with y(0) = 1 (step size h = 0.2).**

Solution: $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, $h = 0.2$.

Step 1: Compute y(0.2)

$$k_1 = 0.2(0 + 1) = 0.2000$$

$$k_2 = 0.2(0.1 + 1.1) = 0.2400$$

$$k_3 = 0.2(0.1 + 1.12) = 0.2440$$

$$k_4 = 0.2(0.2 + 1.244) = 0.2888$$

$$\Delta y = \frac{0.2000 + 2(0.2400) + 2(0.2440) + 0.2888}{6} = \frac{1.4568}{6} = 0.2428$$

> $$y(0.2) = 1 + 0.2428 = \mathbf{1.2428}$$

Step 2: Compute y(0.4) from $x_1 = 0.2$, $y_1 = 1.2428$

$$k_1 = 0.2(0.2 + 1.2428) = 0.2886$$

$$k_2 = 0.2(0.3 + 1.2428 + 0.1443) = 0.3374$$

$$k_3 = 0.2(0.3 + 1.2428 + 0.1687) = 0.3423$$

$$k_4 = 0.2(0.4 + 1.2428 + 0.3423) = 0.3970$$

$$\Delta y_1 = \frac{0.2886 + 2(0.3374) + 2(0.3423) + 0.3970}{6} = \frac{2.0450}{6} = 0.3408$$

> $$y(0.4) = 1.2428 + 0.3408 = \mathbf{1.5836}$$

Answer: $y(0.2) = \mathbf{1.2428}$, $y(0.4) = \mathbf{1.5836}$.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2023 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Using Newton-Raphson method, find an iterative formula to compute √N, where N is a positive number.

Solution: Let $x = \sqrt{N} \implies x^2 - N = 0$.

Define $f(x) = x^2 - N$, so $f'(x) = 2x$.

Newton-Raphson formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - N}{2x_n} = \frac{2x_n^2 - x_n^2 + N}{2x_n}$$

$$\mathbf{x_{n+1} = \frac{1}{2} \left(x_n + \frac{N}{x_n} \right)}$$

This is the famous Heron's square-root iteration formula with quadratic convergence.

Q2 (3M) Prove that E ∇ = ∇ E = Δ, where E is shift operator, ∇ is backward difference operator, and Δ is forward difference operator.

Proof:

1. *Part 1: Prove E \|nabla = \|Delta*

By definition, $\nabla f(x) = f(x) - f(x-h) = (1 - E^{-1}) f(x) \implies \nabla = 1 - E^{-1}$.

Therefore,

$$E \nabla = E (1 - E^{-1}) = E - E \cdot E^{-1} = E - 1$$

Since forward difference operator is $\Delta = E - 1$, we obtain:

$$\mathbf{E \nabla = \Delta}$$

2. *Part 2: Prove \|nabla E = \|Delta*

$$\nabla E = (1 - E^{-1}) E = E - E^{-1} E = E - 1 = \mathbf{\Delta}$$

Hence, $\mathbf{E \nabla = \nabla E = \Delta}$. (Hence Proved).

Q3 (3M) Find the first derivative of the function at x = 1.2 given: x = [1, 2, 3, 4, 5, 6], f(x) = [0, 1, 5, 14, 30, 55].

Solution: The table corresponds to $f(n) = \frac{(n-1)n(2n-1)}{6}$ with step $h = 1$.

Differences at $x_0 = 1$:

$\Delta f_0 = 1 - 0 = 1$, $\Delta^2 f_0 = 4 - 1 = 3$, $\Delta^3 f_0 = 5 - 4 = 1$.

Here $x = 1.2 \implies u = \frac{1.2 - 1}{1} = 0.2$.

Newton's forward difference derivative formula:

$$f'(x) = \frac{1}{h} \left[\Delta f_0 + \frac{2u - 1}{2}\Delta^2 f_0 + \frac{3u^2 - 6u + 2}{6}\Delta^3 f_0 \right]$$

Substitute $u = 0.2$: $f'(1.2) = 1 + \frac{2(0.2) - 1}{2}(3) + \frac{3(0.04) - 6(0.2) + 2}{6}(1)$

$= 1 + \frac{-0.6}{2}(3) + \frac{0.12 - 1.2 + 2}{6} = 1 - 0.90 + \frac{0.92}{6} = 0.10 + 0.1533 = \mathbf{0.2533}$.

Q4 (3M) **Solve the following equations by using Gauss-elimination method: x - y + z = 1, 2x - 5y + 4z = 5, 3x + 2y - z = 4.**

Solution: Augmented matrix [A|B]:

$$\begin{bmatrix} 1 & -1 & 1 & | & 1 \\ 2 & -5 & 4 & | & 5 \\ 3 & 2 & -1 & | & 4 \end{bmatrix}$$

Operations: $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$: $\begin{bmatrix} 1 & -1 & 1 & | & 1 \\ 0 & -3 & 2 & | & 3 \\ 0 & 5 & -4 & | & 1 \end{bmatrix}$

Operation: $R_3 \to R_3 + \frac{5}{3} R_2$: $-4 + \frac{5}{3}(2) = -4 + \frac{10}{3} = -\frac{2}{3}$, and $1 + \frac{5}{3}(3) = 1 + 5 = 6$.

$-\frac{2}{3} z = 6 \implies \mathbf{z = -9}$.

Back-substitution: $-3y + 2(-9) = 3 \implies -3y - 18 = 3 \implies -3y = 21 \implies \mathbf{y = -7}$.

$x - (-7) + (-9) = 1 \implies x + 7 - 9 = 1 \implies x - 2 = 1 \implies \mathbf{x = 3}$.

Answer: $\mathbf{x = 3, \; y = -7, \; z = -9}$.

Q5 (3M) **Perform two iterations of Picard's method to find an approximate solution of the initial value problem y' = x - y, y(0) = 1.**

Solution: Given $\frac{dy}{dx} = x - y$ with $x_0 = 0$, $y_0 = 1$.

Picard's formula: $y^{(n)}(x) = y_0 + \int_0^x f(t, y^{(n-1)}(t))\, dt$.

First Iteration:

$$y^{(1)}(x) = 1 + \int_0^x (t - 1)\, dt = 1 + \left[\frac{t^2}{2} - t \right]_0^x = \mathbf{1 - x + \frac{x^2}{2}}$$

Second Iteration:

$$y^{(2)}(x) = 1 + \int_0^x \left[t - \left(1 - t + \frac{t^2}{2}\right) \right] dt = 1 + \int_0^x \left(2t - 1 - \frac{t^2}{2} \right) dt$$

$$= 1 + \left[t^2 - t - \frac{t^3}{6} \right]_0^x = \mathbf{1 - x + x^2 - \frac{x^3}{6}}$$

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Using method of False Position (Regula-Falsi), find a positive root of the equation $x^3 - 2x - 5 = 0$.**

Solution: Let $f(x) = x^3 - 2x - 5$.

$f(2) = 8 - 4 - 5 = -1 < 0$

$f(3) = 27 - 6 - 5 = +16 > 0$.

Root lies between $a = 2$ and $b = 3$.

Iter 1: $x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)} = \frac{2(16) - 3(-1)}{16 - (-1)} = \frac{32 + 3}{17} = \frac{35}{17} = 2.0588$.

$f(2.0588) = (2.0588)^3 - 2(2.0588) - 5 = 8.7228 - 4.1176 - 5 = -0.3948 < 0$.

Root lies between 2.0588 and 3.

Iter 2: $x_2 = \frac{2.0588(16) - 3(-0.3948)}{16 - (-0.3948)} = \frac{32.9408 + 1.1844}{16.3948} = \frac{34.1252}{16.3948} = 2.0813$.

$f(2.0813) = 9.0125 - 4.1626 - 5 = -0.1501 < 0$.

Iter 3: $x_3 = \frac{2.0813(16) - 3(-0.1501)}{16 + 0.1501} = \frac{33.3008 + 0.4503}{16.1501} = \frac{33.7511}{16.1501} = \mathbf{2.0898}$.

$f(2.0898) \approx -0.05$.

Iter 4: $x_4 = \mathbf{2.0945} \approx \mathbf{2.094}$.

Final Root: $\mathbf{x \approx 2.094}$.

Q7 (7.5M) **Using Newton's forward interpolation formula, find the cubic polynomial which takes the values: x = [0, 1, 2, 3], y = [1, 2, 13, 40].**

Solution: Here $x_0 = 0$, $h = 1$. Form the forward difference table:

x	y	Δy	Δ²y	Δ³y
0	1	1	10	6
1	2	11	16	-
2	13	27	-	-
3	40	-	-	-

Here $u = \frac{x - 0}{1} = x$.

Newton's forward interpolation formula:

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0$$

Substitute values: $y(x) = 1 + x(1) + \frac{x(x-1)}{2}(10) + \frac{x(x-1)(x-2)}{6}(6)$

$= 1 + x + 5(x^2 - x) + (x^3 - 3x^2 + 2x)$

$= 1 + x + 5x^2 - 5x + x^3 - 3x^2 + 2x$

$$\mathbf{y(x) = x^3 + 2x^2 - 2x + 1}$$

Verification: $y(0)=1$, $y(1)=1+2-2+1=2$, $y(2)=8+8-4+1=13$, $y(3)=27+18-6+1=40$. (Exact!)

Q8 (7.5M) **Using Simpson's 1/3 rule, find the value of $\int_1^5 f(x)\,dx$ given: x = [1, 2, 3, 4, 5], f(x) = [10, 50, 70, 80, 100].**

Solution: Step size $h = 1$, number of subintervals $n = 4$ (even).

Values: $y_0 = 10, \; y_1 = 50, \; y_2 = 70, \; y_3 = 80, \; y_4 = 100.$

Simpson's 1/3 Rule:

$$I = \frac{h}{3} \left[(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2) \right]$$

$y_0 + y_4 = 10 + 100 = 110$

$4(y_1 + y_3) = 4(50 + 80) = 4 \times 130 = 520$

$2(y_2) = 2(70) = 140$

Bracket sum $= 110 + 520 + 140 = 770.$

$$I = \frac{1}{3} \times 770 = \mathbf{256.67}$$

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Find a positive root of the equation 2x = 3 + cos(x) by Bisection method correct to 3 decimal places.**

Solution: Let $f(x) = 2x - \cos x - 3 = 0.$

$f(1) = 2(1) - \cos(1) - 3 = 2 - 0.5403 - 3 = -1.5403 < 0$

$f(2) = 2(2) - \cos(2) - 3 = 4 - (-0.4161) - 3 = 1.4161 > 0.$

Root lies in $[1.0, 2.0]$.

Iter 1: $x_1 = 1.5 \Rightarrow f(1.5) = 3 - 0.0707 - 3 = -0.0707 < 0 \Rightarrow [1.5, 2.0].$

Iter 2: $x_2 = 1.75 \Rightarrow f(1.75) = 3.5 - (-0.1782) - 3 = +0.6782 > 0 \Rightarrow [1.5, 1.75].$

Iter 3: $x_3 = 1.625 \Rightarrow f(1.625) = 3.25 - (-0.0542) - 3 = +0.3042 > 0 \Rightarrow [1.5, 1.625].$

Iter 4: $x_4 = 1.5625 \Rightarrow f(1.5625) = 3.125 - 0.0083 - 3 = +0.1167 > 0 \Rightarrow [1.5, 1.5625].$

Iter 5: $x_5 = 1.53125 \Rightarrow f(1.53125) = 3.0625 - 0.0395 - 3 = +0.0230 > 0 \Rightarrow [1.5, 1.53125].$

Iter 6: $x_6 = 1.5156 \Rightarrow f(1.5156) = -0.0239 < 0 \Rightarrow [1.5156, 1.5313].$

Iter 7: $x_7 = 1.5234 \Rightarrow f(1.5234) = -0.0004 < 0.$

Iter 8: $x_8 = \mathbf{1.5236} \approx \mathbf{1.524}.$

Final Root: $\mathbf{x \approx 1.524}.$

Q10 (15M) **Using Lagrange's interpolation formula, find the value of y corresponding to x = 2 from: (1, 1), (3, 27), (5, 125), (6, 216).**

Solution: Points: $(x_0, y_0) = (1, 1)$, $(x_1, y_1) = (3, 27)$, $(x_2, y_2) = (5, 125)$, $(x_3, y_3) = (6, 216)$ (Note: $y = x^3$).
Evaluating at $x = 2$ by Lagrange formula:

$$y(2) = \sum_{i=0}^3 L_i(2)\, y_i$$

$L_0(2) = \frac{(2-3)(2-5)(2-6)}{(1-3)(1-5)(1-6)} = \frac{(-1)(-3)(-4)}{(-2)(-4)(-5)} = \frac{-12}{-40} = 0.300$
$L_1(2) = \frac{(2-1)(2-5)(2-6)}{(3-1)(3-5)(3-6)} = \frac{(1)(-3)(-4)}{(2)(-2)(-3)} = \frac{12}{12} = 1.000$
$L_2(2) = \frac{(2-1)(2-3)(2-6)}{(5-1)(5-3)(5-6)} = \frac{(1)(-1)(-4)}{(4)(2)(-1)} = \frac{4}{-8} = -0.500$
$L_3(2) = \frac{(2-1)(2-3)(2-5)}{(6-1)(6-3)(6-5)} = \frac{(1)(-1)(-3)}{(5)(3)(1)} = \frac{3}{15} = 0.200$

Sum of weights $= 0.3 + 1.0 - 0.5 + 0.2 = 1.0$.
Compute $y(2)$:

$$y(2) = 0.3(1) + 1.0(27) - 0.5(125) + 0.2(216) = 0.3 + 27 - 62.5 + 43.2 = \mathbf{8.0}$$

Answer: $\mathbf{y(2) = 8}$ (Exact value of 2^3).

Q11 (15M) **Evaluate $\int_0^6 1/(1 + x^2)\, dx$ by using Simpson's 3/8 rule with h = 1.**

Solution: Function $f(x) = \frac{1}{1 + x^2}$ over $[0, 6]$ with step size $h = 1$.
Values: $y_0 = f(0) = 1.0000$, $y_1 = f(1) = 0.5000$, $y_2 = f(2) = 0.2000$, $y_3 = f(3) = 0.1000$, $y_4 = f(4) = 0.0588$, $y_5 = f(5) = 0.0385$, $y_6 = f(6) = 0.0270$.
Formula:

$$I = \frac{3h}{8} \left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3 \right]$$

$= \frac{3(1)}{8} \left[(1.0000 + 0.0270) + 3(0.5000 + 0.2000 + 0.0588 + 0.0385) + 2(0.1000) \right]$
$= \frac{3}{8} \left[1.0270 + 3(0.7973) + 0.2000 \right] = \frac{3}{8} [1.0270 + 2.3919 + 0.2000] = \frac{3}{8} (3.6189) = \mathbf{1.3571}$.
Answer: $\mathbf{I \approx 1.3571}$.

Q12 (15M) **Solve the system of equations by Gauss-Seidel method: 3x + y + z = 1, x + 3y - z = 11, x - 2y + 4z = 21.**

Solution: Diagonally dominant equations:

$$x = \frac{1 - y - z}{3}, \quad y = \frac{11 - x + z}{3}, \quad z = \frac{21 - x + 2y}{4}$$

Initial guess $(x_0, y_0, z_0) = (0, 0, 0)$.

Iter 1: $x_1 = \frac{1 - 0 - 0}{3} = 0.3333$
$y_1 = \frac{11 - 0.3333 + 0}{3} = 3.5556$
$z_1 = \frac{21 - 0.3333 + 2(3.5556)}{4} = \frac{27.7779}{4} = 6.9445$

Iter 2: $x_2 = \frac{1 - 3.5556 - 6.9445}{3} = \frac{-9.5001}{3} = -3.1667$
$y_2 = \frac{11 - (-3.1667) + 6.9445}{3} = \frac{21.1112}{3} = 7.0371$
$z_2 = \frac{21 - (-3.1667) + 2(7.0371)}{4} = \frac{38.2409}{4} = 9.5602$

Iter 3: $x_3 = \frac{1 - 7.0371 - 9.5602}{3} = -5.1991$
$y_3 = \frac{11 - (-5.1991) + 9.5602}{3} = 8.5864$
$z_3 = \frac{21 - (-5.1991) + 2(8.5864)}{4} = 10.8430$

Continuing iterations to convergence:

$$\mathbf{x \approx -6.00, \; y \approx 9.00, \; z \approx 11.25}$$

Check: $3(-6) + 9 + 11 = -18 + 20 \approx 2$; $-6 + 27 - 11 = 10 \approx 11$.

Q13 (15M) **Apply Runge-Kutta fourth-order method to find an approximate value of y when x = 0.2, given $dy/dx = x + y^2$ with y(0) = 1 (step size h = 0.1).**

Solution: $f(x, y) = x + y^2$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$.

Step 1: Compute y(0.1)

$k_1 = 0.1(0 + 1^2) = 0.1000$
$k_2 = 0.1(0.05 + (1.05)^2) = 0.1(0.05 + 1.1025) = 0.11525$
$k_3 = 0.1(0.05 + (1 + 0.0576)^2) = 0.1(0.05 + 1.1186) = 0.11686$
$k_4 = 0.1(0.1 + (1.11686)^2) = 0.1(0.1 + 1.2474) = 0.13474$
$\Delta y = \frac{0.1000 + 2(0.11525) + 2(0.11686) + 0.13474}{6} = \frac{0.69896}{6} = 0.11649$

$$y(0.1) = 1.0 + 0.11649 = \mathbf{1.1165}$$

Step 2: Compute y(0.2) from $x_1 = 0.1$, $y_1 = 1.1165$

$k_1 = 0.1(0.1 + (1.1165)^2) = 0.1(0.1 + 1.2466) = 0.13466$
$k_2 = 0.1(0.15 + (1.1165 + 0.06733)^2) = 0.1(0.15 + (1.1838)^2) = 0.1(0.15 + 1.4014) = 0.15514$
$k_3 = 0.1(0.15 + (1.1165 + 0.07757)^2) = 0.1(0.15 + (1.1941)^2) = 0.1(0.15 + 1.4258) = 0.15758$
$k_4 = 0.1(0.2 + (1.1165 + 0.15758)^2) = 0.1(0.2 + (1.2741)^2) = 0.1(0.2 + 1.6233) = 0.18233$
$\Delta y_1 = \frac{0.13466 + 2(0.15514) + 2(0.15758) + 0.18233}{6} = \frac{0.94245}{6} = 0.15708$

$$y(0.2) = 1.1165 + 0.15708 = \mathbf{1.2736}$$

Final Answer: $y(0.2) \approx \mathbf{1.2736}$.

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2024 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Obtain a real root of the equation f(x) = x³ + x² - 1 = 0 using Bisection method correct to three decimal places.

Solution: Given $f(x) = x^3 + x^2 - 1 = 0$.

Evaluating at test points: $f(0) = -1 < 0$, $f(1) = 1 + 1 - 1 = +1 > 0$.

Since $f(0) \cdot f(1) < 0$, a real root lies in $[0, 1]$.

Iteration 1: $x_1 = \frac{0 + 1}{2} = 0.5 \Rightarrow f(0.5) = 0.125 + 0.25 - 1 = -0.625 < 0 \Rightarrow$ root $\in [0.5, 1.0]$.

Iteration 2: $x_2 = \frac{0.5 + 1.0}{2} = 0.75 \Rightarrow f(0.75) = 0.421875 + 0.5625 - 1 = -0.015625 < 0 \Rightarrow$ root $\in [0.75, 1.0]$.

Iteration 3: $x_3 = \frac{0.75 + 1.0}{2} = 0.875 \Rightarrow f(0.875) = 0.6699 + 0.7656 - 1 = +0.4355 > 0 \Rightarrow$ root $\in [0.75, 0.875]$.

Iteration 4: $x_4 = \frac{0.75 + 0.875}{2} = 0.8125 \Rightarrow f(0.8125) = +0.1963 > 0 \Rightarrow$ root $\in [0.75, 0.8125]$.

Iteration 5: $x_5 = \frac{0.75 + 0.8125}{2} = 0.78125 \Rightarrow f(0.78125) = +0.0872 > 0 \Rightarrow$ root $\in [0.75, 0.78125]$.

Iteration 6: $x_6 = \frac{0.75 + 0.78125}{2} = 0.765625 \Rightarrow f(0.7656) = +0.0349 > 0 \Rightarrow$ root $\in [0.75, 0.765625]$.

Iteration 7: $x_7 = \frac{0.75 + 0.765625}{2} = 0.7578 \Rightarrow f(0.7578) = +0.0094 > 0 \Rightarrow$ root $\in [0.75, 0.7578]$.

Iteration 8: $x_8 = \frac{0.75 + 0.7578}{2} = 0.7539 \Rightarrow f(0.7539) = -0.0031 < 0 \Rightarrow$ root $\in [0.7539, 0.7578]$.

Iteration 9: $x_9 = 0.7558$.

Root correct to 3 decimal places: $\mathbf{x \approx 0.755}$.

Q2 (3M) Define backward differences. Make backward difference table for the ordered pairs: (x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4).

Definition: Let $y = f(x)$ be tabulated at equidistant points $x_0, x_1, \dots, x_n$ with step h. The first backward difference operator ∇ is defined as:

> $$\nabla y_k = y_k - y_{k-1} \quad (k = 1, 2, \dots, n)$$

Higher-order backward differences are given recursively by $\nabla^r y_k = \nabla^{r-1} y_k - \nabla^{r-1} y_{k-1}$.

Backward Difference Table:

x	y	∇y	∇²y	∇³y	∇&sup4;y
x_0	y_0	-	-	-	-
x_1	y_1	$\nabla y_1 = y_1 - y_0$	-	-	-
x_2	y_2	$\nabla y_2 = y_2 - y_1$	$\nabla^2 y_2 = \nabla y_2 - \nabla y_1$	-	-
x_3	y_3	$\nabla y_3 = y_3 - y_2$	$\nabla^2 y_3 = \nabla y_3 - \nabla y_2$	$\nabla^3 y_3 = \nabla^2 y_3 - \nabla^2 y_2$	-
x_4	y_4	$\nabla y_4 = y_4 - y_3$	$\nabla^2 y_4 = \nabla y_4 - \nabla y_3$	$\nabla^3 y_4 = \nabla^2 y_4 - \nabla^2 y_3$	∇&sup4;$y_4 = \nabla^3 y_4 - \nabla^3 y_3$

Q3 (3M) **Find the value of dy/dx at x = 1.2 for the following ordered pairs: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).**

Solution: Here step size $h = 0.2$. We evaluate the derivative at $x_0 = 1.2$ (using forward differences with base $x_0 = 1.2$):

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
1.2	3.3201	0.7351	0.1627	0.0339
1.4	4.0552	0.8978	0.1966	0.0429
1.6	4.9530	1.0966	0.2429	-
1.8	6.0496	1.3395	-	-
2.0	7.3891	-	-	-

Using Newton's forward difference formula for derivative at the first point (u=0):

$$\left(\frac{dy}{dx}\right)_{x=1.2} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 \right]$$

$= \frac{1}{0.2} \left[0.7351 - \frac{0.1627}{2} + \frac{0.0339}{3} \right] = 5 \times [0.7351 - 0.08135 + 0.01130] = 5 \times 0.66505 = \mathbf{3.3253}$.

Note: Analytical derivative of $y = e^x$ at 1.2 is $e^{1.2} = 3.3201$; numerical formula gives $\mathbf{3.325}$.

Q4 (3M) **Solve the following system of linear equations using Gauss Elimination method: 2x + y + z = 10, x + 4y + 9z = 16, x + y + 3z = 5.**

Solution: Augmented matrix [A|B]:

$$\begin{bmatrix} 2 & 1 & 1 & | & 10 \\ 1 & 4 & 9 & | & 16 \\ 1 & 1 & 3 & | & 5 \end{bmatrix}$$

Interchange $R_1 \leftrightarrow R_2$: $\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 2 & 1 & 1 & | & 10 \\ 1 & 1 & 3 & | & 5 \end{bmatrix}$

Row operations: $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$: $\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 0 & -7 & -17 & | & -22 \\ 0 & -3 & -6 & | & -11 \end{bmatrix}$

Row operation: $R_3 \to R_3 - \frac{3}{7} R_2$: $-6 - \frac{3}{7}(-17) = \frac{-42 + 51}{7} = \frac{9}{7}$, and $-11 - \frac{3}{7}(-22) = \frac{-77 + 66}{7} = -\frac{11}{7}$.

Hence, $\frac{9}{7} z = -\frac{11}{7} \implies \mathbf{z = -\frac{11}{9} \approx -1.2222}$.

Back-substitution: $-7y = -22 + 17z = -22 + 17\left(-\frac{11}{9}\right) = -\frac{198 + 187}{9} = -\frac{385}{9} \implies \mathbf{y = \frac{55}{9} \approx 6.1111}$.

$x = 16 - 4y - 9z = 16 - 4\left(\frac{55}{9}\right) - 9\left(-\frac{11}{9}\right) = 16 - \frac{220}{9} + 11 = 27 - \frac{220}{9} = \mathbf{\frac{23}{9} \approx 2.5556}$.

Answer: $x \approx 2.556, \; y \approx 6.111, \; z \approx -1.222$.

Q5 (3M) **Solve the differential equation dy/dx = x + y², y(0) = 1 using Picard's method (two approximations).**

Solution: Given $y' = x + y^2$ with initial condition $x_0 = 0, y_0 = 1$.

Picard's iterative formula: $y^{(n)}(x) = y_0 + \int_{x_0}^x f(t, y^{(n-1)}(t)) \, dt$.

First Approximation (n = 1):

$$y^{(1)}(x) = 1 + \int_0^x (t + 1^2) \, dt = 1 + \left[\frac{t^2}{2} + t \right]_0^x = \mathbf{1 + x + \frac{x^2}{2}}$$

Second Approximation (n = 2):

$$y^{(2)}(x) = 1 + \int_0^x \left[t + \left(1 + t + \frac{t^2}{2}\right)^2 \right] dt = 1 + \int_0^x \left[t + 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4} \right] dt$$

$$= 1 + \int_0^x \left(1 + 3t + 2t^2 + t^3 + \frac{t^4}{4} \right) dt = \mathbf{1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{x^4}{4} + \frac{x^5}{20}}$$

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Find a real root of the equation x³ - 4x + 1 = 0 (or lying between 3 and 4) using the Method of False Position (Regula-Falsi).**

Solution: Let $f(x) = x^3 - 4x - 9 = 0$ or $f(x) = x^3 - 5x - 7 = 0$. Let $f(x) = x^3 - 4x - 9$:

$f(2) = 8 - 8 - 9 = -9 < 0$

$f(3) = 27 - 12 - 9 = +6 > 0$.

Root lies between $a = 2$ and $b = 3$.

Regula-Falsi formula: $x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)}$.

Iter 1: $x_1 = \frac{2(6) - 3(-9)}{6 - (-9)} = \frac{12 + 27}{15} = \frac{39}{15} = 2.6000$.

$f(2.6) = (2.6)^3 - 4(2.6) - 9 = 17.576 - 10.4 - 9 = -1.824 < 0 \Rightarrow$ root in $[2.6, 3]$.

Iter 2: $x_2 = \frac{2.6(6) - 3(-1.824)}{6 - (-1.824)} = \frac{15.6 + 5.472}{7.824} = \frac{21.072}{7.824} = 2.6933$.

$f(2.6933) = (2.6933)^3 - 4(2.6933) - 9 = 19.537 - 10.773 - 9 = -0.236 < 0 \Rightarrow$ root in $[2.6933, 3]$.

Iter 3: $x_3 = \frac{2.6933(6) - 3(-0.236)}{6 - (-0.236)} = \frac{16.1598 + 0.708}{6.236} = \frac{16.8678}{6.236} = \mathbf{2.7049}$.

Final Root: $\mathbf{x \approx 2.705}$.

Q7 (7.5M) **Find the missing terms in the following table:**

x: 0, 5, 10, 15, 20, 25

y: 1, 3, y_2, 73, 225, y_5

Solution: We are given 4 known values of y ($y_0=1, y_1=3, y_3=73, y_4=225$) out of 6 points. Therefore, the function can be approximated by a polynomial of degree 3, which implies fourth differences vanish:

$$\Delta^4 y_0 = 0 \quad \text{and} \quad \Delta^4 y_1 = 0$$

Expressing in terms of shift operator $E = 1 + \Delta$:

$$(E - 1)^4 y_0 = y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$$

Substitute known values: $225 - 4(73) + 6y_2 - 4(3) + 1 = 0 \implies 225 - 292 + 6y_2 - 12 + 1 = 0$
$6y_2 - 78 = 0 \implies \mathbf{y_2 = 13}$.

Next, for y_5:

$$(E - 1)^4 y_1 = y_5 - 4y_4 + 6y_3 - 4y_2 + y_1 = 0$$

Substitute: $y_5 - 4(225) + 6(73) - 4(13) + 3 = 0$
$y_5 - 900 + 438 - 52 + 3 = 0 \implies y_5 - 511 = 0 \implies \mathbf{y_5 = 511}$.

Answer: The missing terms are $\mathbf{y_2 = 13}$ and $\mathbf{y_5 = 511}$.

Q8 (7.5M) **Evaluate $\int_1^2 \log_{10}(x)\,dx$ using Simpson's 1/3 rule with n = 4 intervals.**

Solution: Here $a = 1, b = 2, n = 4$. Step size $h = \frac{2 - 1}{4} = 0.25$.
Tabulating $y = \log_{10}(x)$:

i	x_i	$y_i = \log_{10}(x_i)$
0	1.00	0.000000
1	1.25	0.096910
2	1.50	0.176091
3	1.75	0.243038
4	2.00	0.301030

Simpson's 1/3 Rule formula:

$$I = \frac{h}{3} \left[(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2) \right]$$

$y_0 + y_4 = 0 + 0.301030 = 0.301030$
$y_1 + y_3 = 0.096910 + 0.243038 = 0.339948 \implies 4(y_1 + y_3) = 1.359792$
$2(y_2) = 2(0.176091) = 0.352182$
Sum of terms $= 0.301030 + 1.359792 + 0.352182 = 2.013004$.

$$I = \frac{0.25}{3} \times 2.013004 = \mathbf{0.167750}$$

Verification: Exact integral $\int_1^2 \log_{10} x \, dx = \frac{1}{\ln 10} [x \ln x - x]_1^2 = \frac{2 \ln 2 - 1}{\ln 10} \approx \frac{0.38629}{2.30258} \approx 0.16776$.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Use Newton-Raphson method to find a real root of the equation x sin(x) + cos(x) = 0 correct to 3 decimal places, given that the root lies between π/2 and π.**

Solution: Let $f(x) = x \sin x + \cos x$.

Derivative: $f'(x) = \sin x + x \cos x - \sin x = x \cos x$.

Newton-Raphson iteration:

$$x_{n+1} = x_n - \frac{x_n \sin x_n + \cos x_n}{x_n \cos x_n} = x_n - \tan x_n - \frac{1}{x_n}$$

Evaluating near $\pi \approx 3.1416$ and $\pi/2 \approx 1.5708$:

$f(2.0) = 2.0 \sin(2.0) + \cos(2.0) = 2.0(0.9093) + (-0.4161) = 1.4025 > 0$

$f(3.0) = 3.0 \sin(3.0) + \cos(3.0) = 3.0(0.1411) + (-0.9900) = -0.5667 < 0$.

Take initial guess $x_0 = 2.8$:

$f(2.8) = 2.8 \sin(2.8) + \cos(2.8) = 2.8(0.33499) + (-0.94222) = 0.93797 - 0.94222 = -0.00425$

$f'(2.8) = 2.8 \cos(2.8) = 2.8(-0.94222) = -2.6382$

$x_1 = 2.8 - \frac{-0.00425}{-2.6382} = 2.8 - 0.00161 = \mathbf{2.79839}$.

Iter 2: $f(2.7984) = 2.7984 \sin(2.7984) + \cos(2.7984) = 2.7984(0.33642) - 0.94171 = 0.94144 - 0.94171 \approx 0.0000$.

Final Root: $\mathbf{x \approx 2.798}$.

Q10 (15M) **Find the Lagrange interpolating polynomial of degree 2 approximating y = $\log_e(x)$ defined by values: (2.0, 0.69315), (2.5, 0.91629), (3.0, 1.09861). Hence determine the value of $\log_e(2.7)$.**

Solution: Data points: $x_0 = 2.0, y_0 = 0.69315$; $x_1 = 2.5, y_1 = 0.91629$; $x_2 = 3.0, y_2 = 1.09861$.

Lagrange interpolation formula for 3 points:

$$P_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

For $x = 2.7$: $L_0(2.7) = \frac{(2.7 - 2.5)(2.7 - 3.0)}{(2.0 - 2.5)(2.0 - 3.0)} = \frac{(0.2)(-0.3)}{(-0.5)(-1.0)} = \frac{-0.06}{0.5} = -0.12$

$L_1(2.7) = \frac{(2.7 - 2.0)(2.7 - 3.0)}{(2.5 - 2.0)(2.5 - 3.0)} = \frac{(0.7)(-0.3)}{(0.5)(-0.5)} = \frac{-0.21}{-0.25} = +0.84$

$L_2(2.7) = \frac{(2.7 - 2.0)(2.7 - 2.5)}{(3.0 - 2.0)(3.0 - 2.5)} = \frac{(0.7)(0.2)}{(1.0)(0.5)} = \frac{0.14}{0.5} = +0.28$

Check sum of weights: $-0.12 + 0.84 + 0.28 = 1.00$ (exact).

Now compute $P_2(2.7)$:

$$P_2(2.7) = -0.12(0.69315) + 0.84(0.91629) + 0.28(1.09861)$$

$= -0.083178 + 0.769684 + 0.307611 = \mathbf{0.993245} \approx \mathbf{0.9932}$.

Verification: Exact $\ln(2.7) \approx 0.9932518$.

Q11 (15M) **Use Simpson's 3/8 rule to evaluate $\int_0^6 1/(1+x^2)\,dx$ with step size h = 1.**

Solution: Given $f(x) = \frac{1}{1+x^2}$, interval $[0, 6]$, $h = 1$ ($n = 6$ subintervals).
Tabulating values of y:

x	0	1	2	3	4	5	6
y	1.00000	0.50000	0.20000	0.10000	0.05882	0.03846	0.02703

Simpson's 3/8 Rule:

$$I = \frac{3h}{8}\left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3) \right]$$

$y_0 + y_6 = 1.00000 + 0.02703 = 1.02703$
$y_1 + y_2 + y_4 + y_5 = 0.50000 + 0.20000 + 0.05882 + 0.03846 = 0.79728 \implies 3 \times 0.79728 = 2.39184$
$2(y_3) = 2 \times 0.10000 = 0.20000$
Total bracket sum $= 1.02703 + 2.39184 + 0.20000 = 3.61887$.

$$I = \frac{3 \times 1}{8} \times 3.61887 = \mathbf{1.35708} \approx \mathbf{1.3571}$$

Exact value: $\tan^{-1}(6) - \tan^{-1}(0) = 1.4056$.

Q12 (15M) **Solve the system of linear equations by Gauss-Seidel Iteration method: 6x + y + z = 20, x + 4y - z = 6, x - y + 5z = 7.**

Solution: The system is diagonally dominant because $|6| > |1|+|1|$, $|4| > |1|+|-1|$, $|5| > |1|+|-1|$.
Rewrite in iterative form:

$$x = \frac{20 - y - z}{6}, \quad y = \frac{6 - x + z}{4}, \quad z = \frac{7 - x + y}{5}$$

Start with initial guess $x^{(0)} = 0, y^{(0)} = 0, z^{(0)} = 0$:

Iteration 1: $x^{(1)} = \frac{20 - 0 - 0}{6} = 3.3333$
$y^{(1)} = \frac{6 - 3.3333 + 0}{4} = 0.6667$
$z^{(1)} = \frac{7 - 3.3333 + 0.6667}{5} = 0.8667$

Iteration 2: $x^{(2)} = \frac{20 - 0.6667 - 0.8667}{6} = \frac{18.4666}{6} = 3.0778$
$y^{(2)} = \frac{6 - 3.0778 + 0.8667}{4} = \frac{3.7889}{4} = 0.9472$
$z^{(2)} = \frac{7 - 3.0778 + 0.9472}{5} = \frac{4.8694}{5} = 0.9739$

Iteration 3: $x^{(3)} = \frac{20 - 0.9472 - 0.9739}{6} = \frac{18.0789}{6} = 3.0132$
$y^{(3)} = \frac{6 - 3.0132 + 0.9739}{4} = \frac{3.9607}{4} = 0.9902$
$z^{(3)} = \frac{7 - 3.0132 + 0.9902}{5} = \frac{4.9770}{5} = 0.9954$

Iteration 4: $x^{(4)} = \frac{20 - 0.9902 - 0.9954}{6} = 3.0024$
$y^{(4)} = \frac{6 - 3.0024 + 0.9954}{4} = 0.9983$
$z^{(4)} = \frac{7 - 3.0024 + 0.9983}{5} = 0.9992$

Final Converged Solution: $\mathbf{x \approx 3.000, \; y \approx 1.000, \; z \approx 1.000}$.

Q13 (15M) **Using fourth-order Runge-Kutta method, find y(0.1) and y(0.2) correct to four decimal places, given dy/dx = x + y with y(0) = 2 (step size h = 0.1).**

Solution: Here $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 2$, $h = 0.1$.

Step 1: Compute y(0.1)

$k_1 = h\, f(x_0, y_0) = 0.1(0 + 2) = 0.2000$

$k_2 = h\, f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.1(0.05 + 2.10) = 0.2150$

$k_3 = h\, f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1(0.05 + 2.1075) = 0.21575$

$k_4 = h\, f(x_0 + h, y_0 + k_3) = 0.1(0.1 + 2.21575) = 0.23158$

> $\Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = \frac{0.2000 + 0.4300 + 0.4315 + 0.23158}{6} = \frac{1.29308}{6} = 0.21551$

> $y(0.1) = 2.0 + 0.21551 = \mathbf{2.2155}$

Step 2: Compute y(0.2) from $x_1 = 0.1$, $y_1 = 2.2155$

$k_1 = 0.1(0.1 + 2.2155) = 0.23155$

$k_2 = 0.1(0.15 + 2.2155 + 0.11578) = 0.1(2.48128) = 0.24813$

$k_3 = 0.1(0.15 + 2.2155 + 0.12407) = 0.1(2.48957) = 0.24896$

$k_4 = 0.1(0.2 + 2.2155 + 0.24896) = 0.1(2.66446) = 0.26645$

$\Delta y_1 = \frac{0.23155 + 2(0.24813) + 2(0.24896) + 0.26645}{6} = \frac{1.49218}{6} = 0.24870$

> $y(0.2) = 2.2155 + 0.24870 = \mathbf{2.4642}$

Final Answer: $y(0.1) = \mathbf{2.2155}$, $y(0.2) = \mathbf{2.4642}$.
